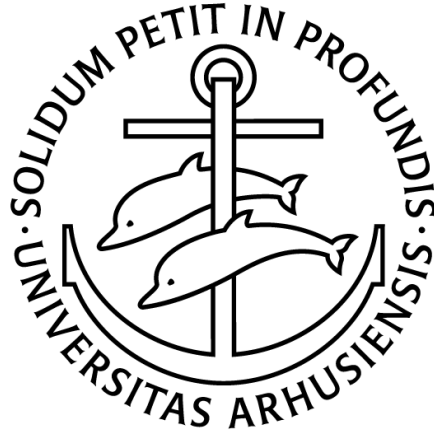




Determinants of Cross-Border Mergers and Acquisitions within the Healthcare Industry: An Empirical Study

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Summary

This paper examines the determinants of cross-border mergers and acquisitions in the healthcare services industry from 2000 to 2024, extending the study of Erel, Liao, and Weisbach (2012). The cross-border M&A literature is built on aggregate, multi-sector samples and explains deal flow through gravitational, financial, and institutional channels, but does not test whether sector-specific regulation is itself a determinant. Healthcare is a natural setting, as licensing and reimbursement regimes make regulatory access an asset a foreign entrant cannot easily replicate. Using Orbis M&A data, I construct a directed bilateral country-pair-year panel of 54,144 observations covering 47 countries, estimated via Poisson pseudo-maximum-likelihood with acquiror, target, and year fixed effects and standard errors clustered at the country-pair level. I test five hypotheses drawn from gravity theory, the Efficient Market Hypothesis, and the Eclectic Paradigm. Consistent with Erel et al. (2012), bilateral trade and geographic proximity significantly shape cross-border deal volume, and the gravity block accounts for the largest share of explanatory power. Legal, religious, and cultural distance each reduce deal flow and together form the largest incremental determinant beyond gravity, consistent with an information-asymmetry interpretation. The financial-arbitrage channels, which in the cross-border M&A literature determine deal flow, cost-of-capital and exchange-rate differences, are not supported in this healthcare sample, only the corporate income tax differential carries a significant effect. Institutional integration raises deal flow through tax treaties and shared OECD membership, but not within the deeply integrated EU, where the relationship reverses. The central result concerns regulation: product market regulation in target-country is the largest coefficient in the model, identified from within-country regulatory change, and its decomposition shows the effect runs specifically through the barriers-to-entry component. The thesis shows that industry matters, in a single heavily regulated sector, it is regulatory access, not financial arbitrage, that drives deal flow.

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1 Introduction

From 2000 to 2024, the global volume of Cross-Border M&A deals grew from ≈ 4400 deals to ≈ 8200 deals, with an average deal value of $\approx$ USD 1 billion in 2000 to $\approx$ USD 540 million in 2024. In comparison, global domestic M&A deal volume grew from $\approx 9,000$ deals in 2000 to over 31,000 deals in 2024, with an average deal value of USD 144 million to $\approx$ USD 52 million in the same period. In this period, out of the total deal volume (domestic and cross-border) cross-border deal volume decreased from 33% to 21%. Cross-border mergers and acquisitions are driven by the same motives as domestic ones: to increase the value of the acquiring firm. Expanding across national borders adds a different layer of complexity and friction: trade barriers, industry regulation, legal system differences, cultural differences, geographic differences, political risks, imperfect capital markets, information asymmetry, exchange rates, and stock market valuations (Erel et al. 2012, p. 1045). Domestic firms operate within a single framework and therefore face less of this friction among firms. How do the cross-border M&A determinants translate into specific industries, such as healthcare? In this paper I replicate the findings of Erel et al. (2012) by extending the sample period from 2000-2024. I add three layers of analysis: industry-level, deal-level and firm-level determinants, testing if the macro determinants hold when controlling for variables in their respective levels of analysis. I sample 54,144 directed bilateral observations, which include country-pair-years with and without cross-border deals. I estimate the extent to which each determinant influences the count of cross-border deals within healthcare services, hence the results are associational rather than causal. In the headline specification I find bilateral trade volume, geographical distance, religious distance, legal distance and corporate income tax differential significant across all specifications consistent with Erel et al. (2012). I find weak significance in total GDP of target country and no significance in the total GDP of acquiror country. In contrast to Erel et al. (2012) I find no significance in bond yield differential. Lastly, in the healthcare specification I find product market regulation significant, while other healthcare variables are insignificant. Decomposing PMR sub-indices, I find barriers to entry significant at the 1% level and state distortions insignificant.

1.1 Healthcare Industry

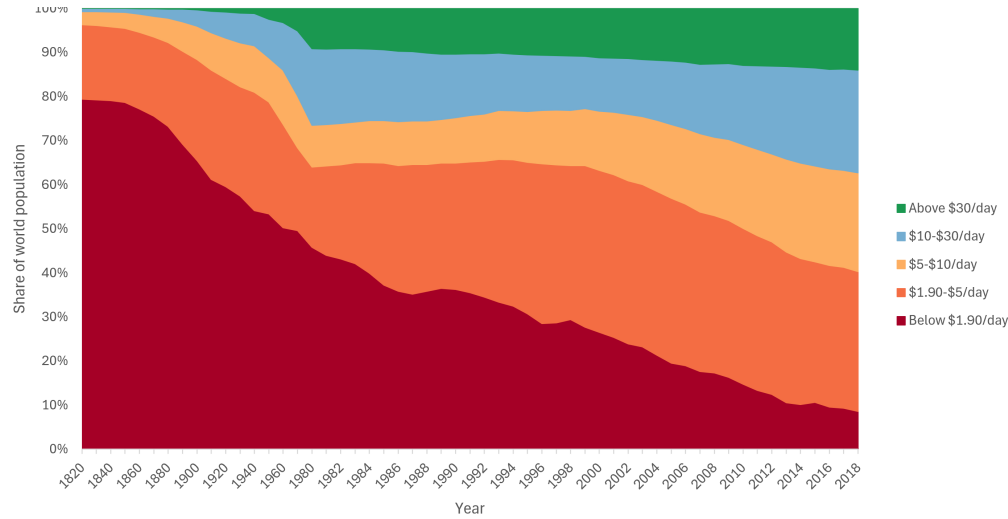
1.1.1 Definition and Scope

The broader healthcare industry covers chemical manufacturing (325), miscellaneous manufacturing (339), health and personal care stores (446), professional, scientific, and technical services (541), ambulatory health care services (621), hospitals (622), nursing and residential care (623). Deal volume and deal values are significantly higher on average within pharmaceuticals and medical device manufacturing. Excluding those subsectors establishes a cleaner dataset with less variance and bias in observed values. Going forward, this thesis focuses exclusively on firms operating in ambulatory health care services (621), hospitals (622), and nursing and residential care (623).

1.1.2 Drivers of Healthcare Demand and industry trends

Grossman (1972) argues the demand for healthcare is determined by the demand for "good health", where an individual is born with an initial stock of health that depreciates over time, at an increasing rate at a certain stage in life, and can be increased by investment (Grossman 1972, pp. 4–7). Consumers therefore demand "good health," not healthcare services. Microeconomic factors that generally predetermines demand for consumption goods (income, prices of goods, and individual preferences), are considered an explanation for the rise in healthcare demand. This associates healthcare demand drivers with the economic development of countries; more developed countries tend to have higher healthcare expenditures as a percentage of GDP. In 1950 the percentage of the global population living in extreme poverty was 53.3%, in 1990 this number was 36.2%, and 8.6% in 2018 (Figure 1). This trend suggest the global living standard has increased significantly, resulting in higher consumer demand and therefore an increasing demand of "good health" utilizing healthcare services. Population demographics data from World Development Indicators Databank displays the average life expectancy by world region from 1950-2023. Average life expectancy rose across every world region from 1950 to 2023 (e.g., 48.9 to 76.6 in South America, 37.4 to 69.7 in Africa). In the same period, the number of age related causes of death, in global top 20 causes of deaths, has increased by 33.3%, while non-age related causes of deaths has decreased by 40%. Based on WHO numbers, 73.4% of COVID-19 deaths from 2020-2023 are associated with elderly people at 65 and over. Appendix A displays top 20 global causes of

Figure 1: global distribution of the worlds population in percentage, living below or above poverty lines



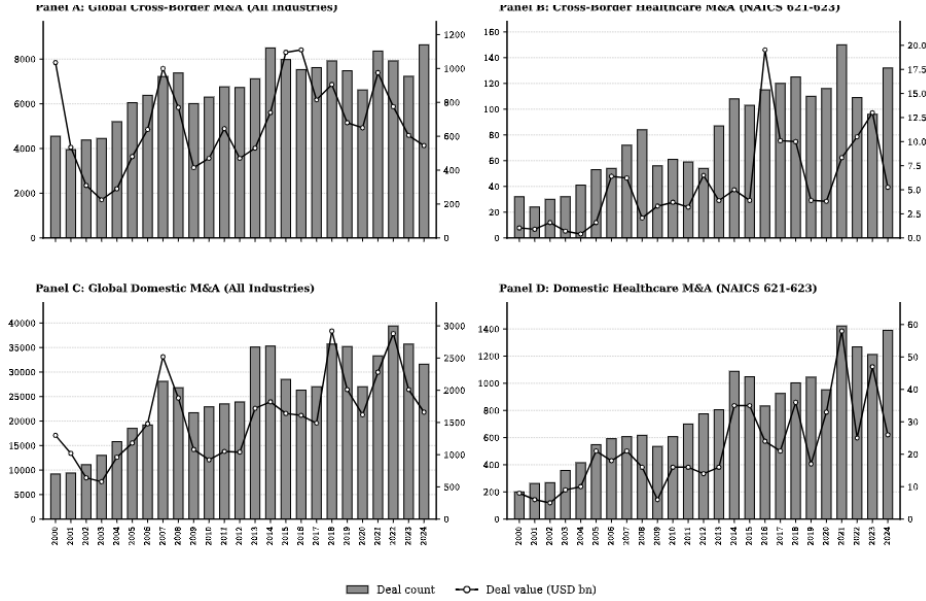
Notes: Own Contribution

death in 2000 and 2021. Moreover, Banerjee (2015) establishes that age is associated with multimorbidity and explains 65% of hospital admissions in the UK are people aged 65 and older (Banerjee 2015, p. 587). Since 2020, healthcare expenditure has spiked across developed regions (North America, Europe, and Asia), mostly associated with COVID-19. Throughout 2023 a sustained increase in healthcare expenditure ocured in North America and Asia, while in Europe it flattens. Alongside demographics, digitalization is reshaping the sector. Toni and Mattia (2022) document a shift toward integrated data systems and remote care, driven by fragmentation, weak interconnection, and pressure to contain costs (Toni and Mattia 2022, p. 5). This structural trend reinforces the long-run rise in healthcare expenditure that, with aging populations, drives demand for healthcare services. Europe, North America, and some places in Asia have higher healthcare data quality, therefore a large part of the healthcare data is skewed towards developed economies. With this information in mind, going forward I assume healthcare demand is driven by two main factors; populations in high GDP economies and age.

1.1.3 Facts and Figures

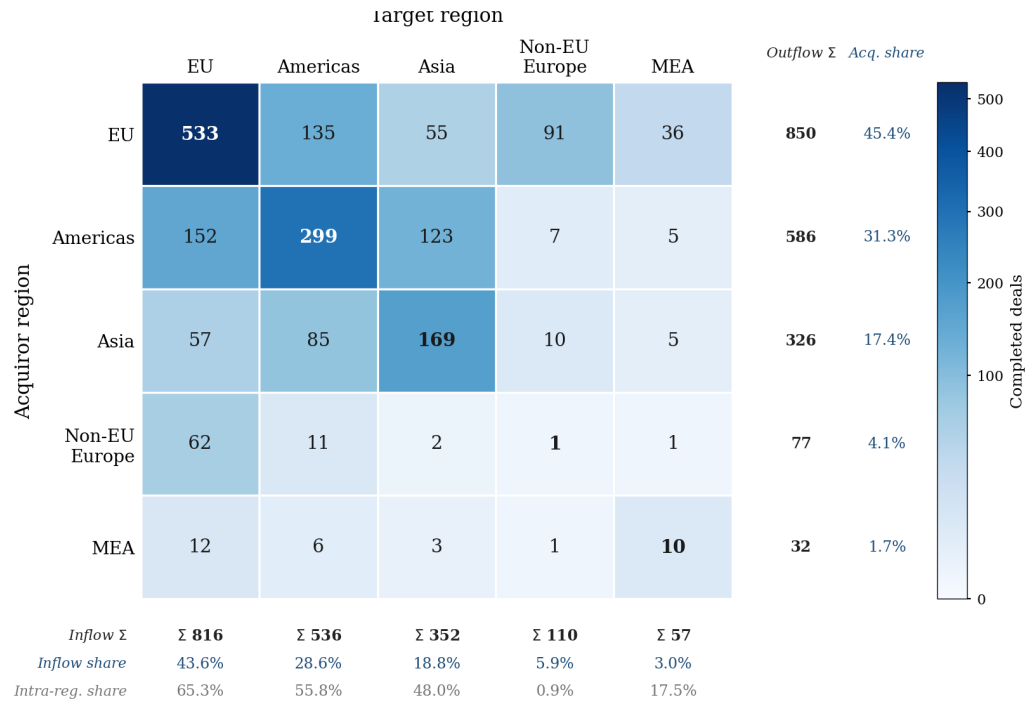
The number of cross-border healthcare M&A deals grew from 32 in 2000 to 132 in 2024, the total deal value grew from USD 1.04bn in 2000 to USD 5.26bn in 2024. The pattern shows a spike in deal volume around the global financial crisis in 2007 and 2008, where 72 deals occurred with a total value of USD 6.22bn and 84 deals occurred with total deal value of USD 2.07bn respectively. From 2013 and onward the average number of deals increased significantly, the total deal value decreased from USD 6.5bn in 2012 to USD 3.88bn in 2013. In 2016 the deal volume increased to 115 with a total deal value of USD 19.53bn. In 2021 deal volume spiked to 150 deals with a total deal value of USD 8.33bn. The deal count rises steadily over the sample period, whereas the deal-value is dominated by large individual transactions. This motivates the count of deals, rather than aggregate deal value, as the dependent variable going forward.

Figure 2: M&A Activity - All industries vs Healthcare (2000-2024)



Notes: Source: Own Contribution; original figure from Erel et al. (2012) is displayed in Appendix C. Panel A displays the number of total cross-border deals within the healthcare industry; the left axis represents the total count, while the right axis shows total deal values in USD millions.

In the sample period, countries in the EU were responsible for 45.4% of the total number of cross-border deals, North America was responsible for 31.3%, Asian countries were responsible for 17.70%, while Non-EU countries accounted for 4.1%, Africa and the Middle East accounted for 1.7% (Figure 3). Cross-border deals between EU countries constituted 65.3% of the EU's total deal volume, cross-border deals between North American countries constituted 55.8% of their total deal volume, cross-border deals between Asian countries constituted 48.0% of their total deal volume, cross-border deals between non-EU European countries constituted 0.9% of their total deal volume, and cross-border deals between African and Middle Eastern countries constituted 17.5% of their respective deal volumes.



Total deals: 1.871 | Cross-border: 859 (45.9%) | Intra-regional: 1.012 (54.1%)

Figure 3: Notes: Cross-border healthcare M&A deal flows by region, cumulative 2000-2024. Cells report completed deals from each acquiror region (rows) to each target region (columns); diagonal cells (bold) are intra-regional. Right margin gives total outflow; the lower panel gives total inflow, each target region's share of all deals, and its intra-regional share (diagonal/inflow). Source: Own contribution.

Overall, the regional clustering suggests firms in the healthcare services industry tend to acquire other firms in nearby countries, and deal volume flows between developed economies and from developed countries into less developed countries. Following Erel et al. (2012) I reconstruct a deal matrix, which reports the bilateral deal-count. Acquiror countries are listed across columns and target countries down rows, diagonal entries are domestic deals and are excluded from the cross-border totals

1.1.4 Industry Regulations

The broader Healthcare Industry is highly regulated, drug and medical device approval is the most visible regulatory layer, healthcare firms in the United States are regulated by the Food and Drug Administration (FDA), and the European Union has moved closest to a harmonised cross-border regime through the European Medicines Agency (EMA). This layer governs products that are mobile and tradable, pharmaceuticals and medical devices, which are the subsectors I exclude from the data sample. Healthcare services are a different layer. No global regulatory regime exists, and the industry is fragmented by country. Within the national layer, healthcare services materialize as hospitals, ambulances, nursing, pre-hospital services, etc., all of which require local facilities and local personnel, but also licensing to operate. Because licenses, accreditation, ownership approvals, and payer relationships are slow and costly to assemble organically, acquiring an incumbent that already holds them is the natural mode of entry. Some countries limit foreign ownership of healthcare service providers, such as Spain and the US, foreign firms can't simply acquire other firms and consolidate industries across borders. Moreover, reimbursement systems also differ from country to country, single payer, social-insurance, and multi-payer, adding another layer of administrative complexity. The regulatory landscape for healthcare firms is complex and creates high entry barriers for foreign firms to enter the market organically, therefore the natural market entry is via acquisitions of national incumbents.

1.2 Research Objective and Question

Fewer people live in extreme poverty, populations are aging worldwide, and economies continue to grow. As a result, demand for healthcare services is rising and the volume of cross-border M&A has increased. This thesis

asks: What determines the volume and direction of cross-border M&A in the healthcare service industry, and do the channels established for cross-border M&A literature (Erel et al. 2012), hold in healthcare, or is it instead determined by regulatory factors?

1.3 Thesis Roadmap

Section 1 defines the healthcare industry and the scope of analysis, presents the facts of cross-border healthcare M&A, and outlines the regulatory fragmentation that motivates the industry specific focus. The remainder of the thesis proceeds as follows. Section 2 reviews the cross-border M&A literature, organized around the determinants of deal flow, the gravity framework, and institutional and cultural distance, and identifies the gap addressed in this thesis. Section 3 develops the theoretical framework, based on neoclassical and new trade theory, the Eclectic Paradigm, the Resource-Based View, and the Efficient Market Hypothesis, and derives five hypotheses (H_1-H_5) tested in the analysis. Section 4 describes the data process, the construction of the directed bilateral panel, the variable definitions, and the summary statistics. Section 5 outlines the empirical strategy, the PPML baseline specification, the identification strategy, and the robustness panels. Section 6 shows the results: the headline regression cascade, the deal and firm-level regressions, the healthcare channel decomposition, and the robustness checks. Section 7 discusses the findings linking them hypothesis by hypothesis and positions them in relation to the existing literature. Section 8 addresses the limitations of this study, Section 9 concludes, and Section 10 suggests directions for future research.

2 Literature Review

In this section, I outline the existing literature on the determinants of cross-border M&A, identify the gap within it, and position this thesis and its contribution. I organise the literature into three strands: the determinants of cross-border M&A, the gravity framework as the empirical foundation for bilateral flows, and institutional and cultural distance as a source of friction. **The determinants of cross-border M&A** The closest literature to this thesis examines what drives the volume and direction of cross-border deal flow. Erel et al. (2012) provide the central reference: using a large multi-country,

multi-sector sample, they show that cross-border M&A is driven by bilateral economic linkage and by financial and valuation factors, with acquirors in more highly-valued markets acquiring targets in less highly-valued ones. Giovanni (2005) reaches a complementary conclusion, showing that cross-border M&A activity, the dominant component of international capital flows, responds to financial-market development and to the cost of financing. Froot and Stein (1991) provide the underlying logic for the valuation channel, arguing that imperfect capital markets allow exchange-rate and wealth differences to translate into foreign acquisitions. The shared finding across this strand is that cross-border M&A is, a financial and gravitational phenomenon. The shared limitation is that these studies pool many sectors into a single aggregate sample, sector-specific determinants are, by construction, averaged out. **The gravity framework.** A second strand provides the empirical structure within which bilateral flows are modelled. The gravity equation, introduced for trade by Tinbergen (1962) and given a theoretical foundation by Anderson (1979) and Anderson and Wincoop (2003), relates bilateral flows to economic mass and to the frictions separating a country pair. Head and Mayer (2014) document its application as the standard toolkit for bilateral flows, and Silva and Tenreyro (2006) establish that such flows, which contain many zero observations, should be estimated by Poisson pseudo-maximum-likelihood rather than by log-linear regression. This strand provides the estimation model framework used in this thesis. It is well established in trade and, increasingly, in M&A, but it has been used to aggregate all-sector M&A flows. **Institutional and cultural distance.** A third strand establishes that the friction term in the gravity framework is not only geographic. Porta et al. (1997) and Porta et al. (1998) show that legal origin and the strength of investor protection shape financial-market development and the cost of investing across borders. Stulz and Williamson (2003) extend this to religion and culture, showing that cultural distance predicts the patterns of foreign investment. Kogut and Singh (1988) provides the composite cultural-distance measure used to operationalize this friction, and Dow and Karunaratna (2006) extends it into a "multidimensional measure of psychic distance" adding language, religion, and political system (Dow and Karunaratna 2006, p. 16). The consistent finding is that legal, institutional, and cultural distance deter cross-border M&A by raising the cost of investments through information gaps between countries. **The gap.** Across all three strands, the cross-border M&A literature is built on aggregate, multi-sector samples and explains deal flow through gravitational, financial, cultural, and institutional-distance channels.

What this literature does not address is whether sector-specific regulation is itself a determinant of cross-border M&A. The healthcare industry is a natural industry to ask this question, because it is among the most heavily regulated sectors, and because drug approval, provider licensing, and reimbursement regimes make regulatory access an asset that a foreign entrant cannot easily replicate. I apply the gravity framework of Erel et al. (2012) to a directed bilateral panel of cross-border healthcare M&A transactions and test product market regulation as a determinant. The contribution is to show that, single sector regulatory barriers to entry, are a significant determinant. Moreover, do the established determinants, financial arbitrage, cultural, and institutional channels, stay significant as determinants of cross-border deal flow?

3 Theory

This chapter covers theories used to estimate the independent variables and their respective mechanisms. I cover four theories, neoclassical and new trade theory, eclectic paradigm, resource-based view and efficient market hypothesis. I will link each variable to its respective theory, and explain its theoretical interpretation and expected coefficient sign, further information regarding variables can be found in Appendix B.

3.1 Neoclassical and New Trade Theory

This section draws on three strands, factor-mobility arbitrage (Mundell 1957; Krugman 1980), gravity equation tradition (Anderson and Wincoop 2003) and firm selection mechanism (Melitz 2003).

Factor-price arbitrage Robert A. Mundell (1957) introduced the theory of *international trade and factor mobility* building on previous trade theories.¹ He suggests international trade happens when e.g. *country_i* is labor abundant and *country_j* is capital abundant, thus creating a comparative advantage for both countries. Hence factor advantages differs between countries and therefore are the underlying drivers of capital flows. Mundell (1957) claims, under his assumptions, that goods are perfect substitutes for capital and labor. Thus free trade in goods will produce the same outcome as free movement

¹Eli Heckscher (1919), Bertil Ohlin (1933), Wolfgang Stolper and Samuelson (1941).

of capital and labor, because both forces push toward the same equilibrium, where prices and factor returns (capital and labor) are equal everywhere (Mundell 1957, p. 321). Within i capital and labor has perfect mobility, between i and j capital is highly mobile where labor is assumed to be partially mobile, where commodities are restricted by tariffs, factors of production, will flow accross borders to equalize their marginal returns through other channels (Mundell 1957, p. 322). Given this framework, Mundell claims goods and production factors are perfect substitutes, thus trade barriers will stimulate mobility of production factors and the same trade flow is realized through other channels. Cross-border M&A is a channel, where capital flows are realized cross borders, by transferring control of productive assets to foreign acquirors. This framework establishes two predictions, first bilateral differences in cost of capital - bond yield difference, stock market difference and FX volatility - should drive cross-border M&A direction, where capital is flowing from abundant low-yield economies to scarce high-yield economies. Second, institutional barriers to trade or investment, stimulate factor movements, which restore the push toward factor-price equalization through a different channel, such as cross-border M&A (Mundell 1957, pp. 323–325). Hence Mundell assumes markets constantly pushes towards perfect competition by firm scaling their operations cross-borders and exploiting arbitrage opportunities. A limitation of Neoclassical Trade theory is the assumption that countries trade with each other because of comparative advantages, thus similar countries do not have comparative advantages, which is a contradiction given trade between US and UK, in my sample, have the highest trade intensity.

New Trade theory Where neoclassical trade theory stops, new trade theory begins. Mundell (1957) claimed trade is driven by comparative advantages, where Paul Krugman (1980) claimed trade is driven by economies of scale, which is internal to firms. Because of the scale economies, markets are imperfectly competitive. Thus trade happens between countries with identical tastes, technology and production factors (Krugman 1980, pp. 469–470). Krugman (1980) explains that is what happens when an economy goes from closed to open. Neoclassical trade theory claims markets return to perfect competition yielding constant returns, where new trade theory claims markets return to perfect monopolistic competition yielding increasing returns, since increasing returns to scale eventually leads to one firm dominating the market.

In an open economy with free competition, customers prefer variety and firms can enter markets freely. This creates a downward-sloping demand curve and drives profits to zero in equilibrium, resulting in a market with many small firms each producing a unique product, each earning just enough to cover fixed cost (Krugman 1980, pp. 472–473). When trade opens between countries at zero transportation cost, wages and the cost of producing a good in either country, will be equal. Thus there will be an increase in the scale of production and in the number of goods available for consumption (Krugman 1980, p. 476). **Selection Mechanism** Melitz (2003) extends Krugman's (1980) paper by adapting a dynamic industry model to monopolistic competition in a general equilibrium setting. Melitz (2003) shows firms with different productivity levels coexist in an industry because each firm faces initial uncertainty concerning its productivity before making an irreversible investment to enter the industry. Entry into the export market is also costly, but occurs after a firm gains knowledge of its productivity (Melitz 2003, p. 1695). Melitz (2003) concludes that least productive firms serve domestic markets, medium productive firms serve foreign markets via export, and most productive firms undertake foreign direct investments (FDI), such as cross-border M&A. Additionally, distance and trade friction reduces number of deals between country pairs, but increases the average value of deals, friction acts as fixed costs, that only high value deals overcome. **Gravity tradition** The gravity equation relates bilateral trade flows to GDP, distance and other factors affecting trade barriers. Anderson (2003) introduces the concept Multilateral resistance, the more resistant a country is to trade with all other countries, the more a country is pushed to trade with a bilateral partner. Anderson concludes that national borders reduce trade between industrialized countries by 20-50% (Anderson and Wincoop 2003, p. 170). Therefore, gravitational mechanisms, like GDP and geographical distance, determine trade between country pairs. Neoclassical and New Trade Theory accounts for many of the bilateral determinants I test, yet it does not explain the firm-level and industry-specific variation captured in the regressions. To explain the firm-level and industry specific motives I apply the Eclectic Paradigm and Ressource-based View frameworks, to illuminate industry and firm determinants. Healthcare services require capital investments and labor intensity, given that healthcare service operations require patient transportation and patient care.

3.2 Eclectic Paradigm Theory

This section covers the three tenets of eclectic paradigm framework; ownership, location and internalization advantages extended by Markusen (1995).

James R. Markusen (1995) *The boundaries of Multinational Enterprises and the Theory of International Trade* extends the *Eclectic Paradigm* framework presented by John H. Dunning in 1976 and 1988. Markusen documents that there are large differences across industries in the degree to which production and sales are accounted for by multinational firms. Multinationals tend to be important in industries and firms with four characteristics: high levels of R&D relative to sales, a large share of professional and technical workers in their workforces, products that are new and/or technically complex, and high level of product differentiation and advertising. Corporate age is highly correlated with multinationality. Trade barriers and transport costs shift activity toward direct investment. From this micro perspective, multinationals are important in industries in which intangible, firm-specific assets are important. These assets can generally be characterized as "knowledge capital," ranging from proprietary product or process know-how to reputations and trademarks (Markusen 1995, p. 173). **Ownership (O) Advantages** If foreign multinational enterprises are exactly identical to domestic firms, they will not find it profitable to enter the domestic market. Because there are added costs of entering markets in another country, these include communication and transport costs, higher costs of stationing personnel abroad, and barriers such as language, customs, and being outside local business and government networks. The multinational enterprise must, therefore, arise because it has a special advantage, such as superior technology or lower costs due to economies of scale (Markusen 1995, p. 174). **Location (L) Advantages** The foreign market must offer a location advantage that makes it profitable to produce the product in the foreign country rather than simply produce it at home and export it to the foreign market. Although tariffs, quotas, transport costs, and cheap factor prices are the most obvious sources of location advantages, factors such as access to customers can also be important. Many multinationals operate in service industries (for example, hotels) in which on-site provision of services is an inherent part of their business (Markusen 1995, p. 174). The healthcare service industry shares characteristics of such firms. **Internalization (I) Advantages** The multinational enterprise must have an internalization advantage. If a company has a proprietary product or production process and if, due to tariffs and transport costs, it is advantageous

to produce the product abroad rather than export it, it is still not obvious that the company should set up a foreign subsidiary. One of several alternatives is to license a foreign firm to produce the product or use the production process. Thus the product or process is exploited internally within the firm rather than at arm's length through markets (Markusen 1995, pp. 174–175). To internalize products and production of these, foreign acquisitions solves this obstacle. Additionally, the framework establishes that foreign acquisitions are positively correlated with the age of the acquiror firm (Markusen 1995). These determinants share characteristics with the healthcare service industry firms in the data sample, hence the mechanism will test if these characteristics are consistent with my findings.

3.3 Ressource-Based View Theory

The Resource-Based View (RBV) elaborates on the ownership tenet of the eclectic paradigm, specifying what gives firms transferable competitive advantage. Introduced by Wernerfelt (1984), the RBV opposes Porter's industry structure framework, which locates the source of firm profitability in industry level forces and explains why some industries are more profitable than others. RBV places the source of firm profitability within the internal resources of the firm (Wernerfelt 1984, pp. 178–180). Based on this framework, Jay Barney (1991) introduced the VRIN framework to analyze a firms sustained competitive advantage, across four dimensions; valuable, rare, imitable, non-substitutability. In 1995 Barney changed VRIN to VRIO, substituting non-substitutability with organization. **Value** do a firms resources and capabilities add value by enabling it to exploit opportunities and neutralize threats? **Rare** if a resource is controlled by numerous competing firms, that resource is unlikely to be a competitive advantage for any of them. **Imitability** if competing firms face a cost disadvantage in imitating these resources and capabilities, firms with these special abilities can obtain a sustained competitive advantage. **Organization** to fully realize firm potential, a firm must be organized to exploit its resources and capabilities (Barney 1995, p. 56). The firm resources - tangible assets, intangible assets, capabilities and organisational routines - in the 1995 RBV framework are static resources. Thus Teece et al. (1997) introduces *dynamic capabilities*, defined as the firms ability to adapt its resources to changing environments (Teece et al. 1997) and Anand and Delios (2002) relates RBV to cross-border M&A and claims acquisitions are a mechanism used to exchange capabilities that are otherwise

not possible to efficiently deploy (Anand and Delios 2002, p. 120). The RBV framework establishes firm profitability is driven by internal firm resources, rather than which industry it competes in. If a firm has a rare and valuable resource that is hard to imitate without high costs, with organizations geared to exploit these resources, the firm achieves sustained competitive advantage. The nature of the firm is to increase its profits, by adapting its supply to customer demand, and production to consumption, which is most efficiently achieved via economies of scale (Coase 1937, p. 387). Thus, the natural expansion of the firm, once the firm has achieved a sustained competitive advantage in its domestic market, is to expand beyond its borders. The mechanism is these firms are more likely to engage in cross-border acquisitions, because they have transferable competitive advantages and capacity to integrate foreign firms.

3.4 Efficient market hypothesis and Information Asymmetry

Efficient Market Hypothesis is the assumption that security prices at any time fully reflect all available information (Fama 1970, p. 383). Thus capital will be located to its most efficient use, given security prices are signals with all available information priced in. Fama (1991) extended the original hypothesis to include three levels of information efficiency; weak-form, where security prices reflects historical information, semi-strong form, where security prices reflects all publicly available information, and strong form, where security prices also reflects private information (Fama 1991, pp. 1576–1577). Cross-border M&A deals generally happens at the semi-strong efficiency level, based on historical and public information. Differences in bond yields and equity valuations between acquiror and target countries, have an effective price advantage. Thus capital flows from low-bond yield economies to high-bond yield economies (Froot and Stein 1991). **Information Asymmetry** Legal barriers reflects another asymmetry presented by Porta et al. (1998) discussing the importance of legal rules and enforcement thereof, to protect shareholders and creditors claims. Porta et al. (1998) establishes a legal framework categorizing countries legal into three legal systems; common-law, civil-law, mixed or customary law. Porta et al. (1998) finds that legal origin predicts differences in investor protection, financial market development and hereof corporate governance quality, which leads to systematic differences in

the cost of information acquisition across country pairs (Porta et al. 1998, p. 1131). Stulz and Williamson (2003) extends this argument of institutional distance between country pairs, predicting religion and cultural distance predicts investment patterns of foreign investors (Stulz and Williamson 2003, p. 315). This creates friction in cross-border M&A deals establishing an opposing channel to the efficient market hypothesis, that predicts legal, institutional, religious and cultural distances, reduces cross-border deal flow between country pairs, because bridging the information gap can be a big hurdle to overcome.

3.5 Hypotheses Development

This section links each of the four theories to the hypotheses it motivates and states the expected coefficient sign for each. Table 2 summarises each hypothesis, its variables, and its predicted relationship, the variable definitions are listed in Appendix B. Neoclassical and New Trade Theory motivates two hypotheses. The gravity tradition predicts that bilateral economic mass raises cross-border deal flow and geographic friction reduces it (H_1), the factor-mobility and institutional-integration strands predict that treaty-based and bloc-based reductions in investment friction, together with the depth of the domestic M&A market, raise cross-border deal flow (H_4). The Efficient Market Hypothesis and the information-asymmetry literature motivate two further hypotheses through its two strands: the valuation strand predicts that bilateral differences in the cost of capital and in tax treatment generate arbitrage incentives that direct cross-border deal flow (H_2), while the information-asymmetry strand predicts that legal, institutional, religious, and cultural distance raise the cost of bridging the information gap and therefore reduce cross-border deal flow (H_3). The Eclectic Paradigm motivates the regulatory hypothesis: where product market regulation raises barriers to entry, greenfield entry into a foreign market becomes a location disadvantage, and the multinational enterprise overcomes this disadvantage by internalizing an incumbent through acquisition (H_5). The Resource-Based View is not tested as a separate hypothesis; it provides the interpretive lens for the firm-level analysis in Section 6, where acquiror resources and capabilities are examined as determinants of deal size.

H_1 — **Gravity (Neoclassical and New Trade Theory)**. Cross-border healthcare deal flow between a country pair increases in the economic mass of the acquiror and target countries and decreases in the geographic distance between them. Bilateral trade, as a measure of established economic linkage, is expected to enter positively.

H_2 — **Valuation arbitrage (Efficient Market Hypothesis)**. Bilateral differences in the cost of capital and in corporate taxation generate arbitrage incentives. Capital is expected to flow from low-yield toward high-yield economies, so the bond yield differential and the corporate income tax differential are expected to enter positively, and foreign exchange volatility is expected to affect deal flow as a currency-risk channel.

H_3 — **Institutional and cultural distance (Information Asymmetry)**. Legal, religious, and cultural distance between the acquiror and target countries raise the cost of acquiring and verifying information across the pair. Each distance measure is therefore expected to enter negatively.

H_4 — **Market depth and financing (New Trade Theory, institutional-integration strand)**. Reductions in institutional barriers to investment raise cross-border deal flow. Bilateral investment and double tax treaties, and shared OECD or EU membership, are expected to enter positively, as is the depth of the domestic M&A market, which proxies for the acquiror's access to deal-making capacity and financing.

H_5 — **Regulation (Eclectic Paradigm)**. Where product market regulation in the target country raises barriers to entry, the location disadvantage of greenfield entry is large, and the multinational enterprise internalizes the an incumbent through acquisition. Target-country product market regulation is therefore expected to enter positively, with the effect concentrated in the barriers-to-entry component of the index.

Table 1: Summary of Hypotheses and Variable Definitions

Hypothesis	Measurement	Expected sign
H_1 : Gravity	1. Target country ln(GDP) (<code>tgt_ln_total_gdp</code>)	(+)
	2. Acquiror country ln(GDP) (<code>acq_ln_total_gdp</code>)	(+)
	3. ln(bilateral distance, km) (<code>ln_distance_km</code>)	(-)
	4. ln(bilateral trade) (<code>ln_bilateral_trade</code>)	(+)
H_2 : Valuation arbitrage	1. Bond yield differential, acq-tgt (<code>bond_yield_diff</code>)	(-)
	2. Real stock return differential, acq-tgt (<code>stock_return_diff_real</code>)	(+)
	3. Bilateral FX volatility, 12-month (<code>fx_volatility_12m_annual</code>)	(-)
	4. Bilateral FX ln-change, annual (<code>fx_ln_change_annual</code>)	(+)
	5. Corporate income tax differential, acq-tgt (<code>cit_signed_diff</code>)	(+)
H_3 : Institutional and cultural distance	1. Bilateral legal-origin distance (<code>legal_distance_bilateral</code>)	(-)
	2. Bilateral religious distance (<code>religious_distance</code>)	(-)
	3. Bilateral cultural distance, Kogut-Singh (<code>cultural_distance</code>)	(-)
H_4 : Market depth and financing	1. ln(1 + acquiror-country domestic deals) (<code>ln_acq_domestic_deal_count</code>)	(+/-) ^a
	2. ln(1 + target-country domestic deals) (<code>ln_tgt_domestic_deal_count</code>)	(+)
	3. Both EU member states (<code>both_eu</code>)	(+)
	4. Both OECD member states (<code>both_oecd</code>)	(+)
	5. Bilateral investment treaty in force (<code>BIT_in_force</code>)	(+)
	6. Bilateral double-taxation treaty in force (<code>DTT_in_force</code>)	(+)
H_5 : Regulation	1. Target country product-market regulation index (<code>tgt_pmr_overall</code>)	(+)
	2. Target country health expenditure (% GDP) (<code>tgt_health_exp_pct_gdp</code>)	(+)
	3. Target country healthcare openness index (<code>tgt_hc_openness_index</code>)	(+)
	4. ln(healthcare service revenue ratio, acq/tgt) (<code>ln_hc_svc_revenue_ratio</code>)	(+)
	5. Target healthcare regulatory regime tier (<code>tgt_reg_tier</code>)	(+)
<i>Dependent variable</i>		
Cross-border deal count	Count of completed cross-border healthcare M&A transactions from acquiror country i to target country j in year t (<code>cb_deal_count</code>)	

^a H_4 generates competing predictions on acquiror-country domestic deal activity: a market-depth interpretation predicts (+), since deeper domestic M&A markets reflect financial development and accumulated dealmaking capability; a substitution interpretation predicts (-), since plentiful domestic targets reduce the incentive to pursue foreign acquisitions. The competing predictions are adjudicated empirically. *Sample*: bilateral country-pair-year panel, 2000-2024, 47 countries, excluding offshore financial centres. Dependent variable estimated via PPML with three-way fixed effects (Year + acquiror country + target country) and pair clustering, following Erel, Liao & Weisbach (2012).

4 Data

4.1 Data Processing

Deal Data is extracted from Orbis M&A database, which accounts for total cross-border deals between country pairs, total number of domestic deals, and total cross-border and domestic deals within healthcare services completed from 2000 to 2024. The sample includes deals from both private and public firms. Data sample consists of cross-border mergers and acquisition deals within the healthcare service industry (NAICS 621, 622, 623). Deals in social assistances, (NAICS 624), are excluded given its number of M&A deals accounts for fewer than 50 of the total deals within this period, with low deal values in comparison. Following Erel et al. (2012) I exclude LBO's, recapitalizations, minority stake purchases $\leq 49\%$ and self-tender offers. I further exclude deals from countries with incomplete trade data, stock market data and foreign exchange rate data between 2000-2024. After exclusions I am left with a sample of 21,531 healthcare service mergers and acquisition deals, covering 47 countries with a total reported transaction value of USD 673.57 billion, 2109 of which are cross-border mergers and acquisition deals, with a total transaction value of USD 136.53 billion. I collect a number of data items from Orbis M&A database including acquiror and target name, ISO2 country codes, deal value, announcement date, date of completion, date of last deal status update, primary industry code, country of domicile. Additionally, I collect data items of deal characteristics when available, like acquired stake in percentage points, method of payment pre-deal multiples of acquiror and target, like EBITDA, EBIT, price-earnings, price-to-book value, return on assets, leverage ratio. I also collect the revenues, shareholder funds, market cap, net profits, enterprise value, number of full time employees, incorporation year, acquiror and target age at deal. Annual bilateral trade data, total import and total export between countries, is extracted from UN Comtrade Database, denominated in USD. GDP data in USD is sourced from World Development Indicators Databank, so is the world governance index, which is dissected in the construction of variables section. To measure the quality of accounting disclosures I use the disclosure index from World Bank Doing Business Protecting Minority Investors index. Following the approach of Erel et al. (2012), I use legal system data sourced from University of Ottawa database juriglobe.com (Porta et al. 1997), culture data sourced from geerthofstede.com (Kogut and Singh 1988), language identification

is based on the language family tree presented by Dow and Karunaratna (2006) and data on religion is sourced from ourworldindata.com (Stulz and Williamson 2003), each variable is elaborated in the variable construction section. I collect the latitudes and longitudes of the capital city of each country from mapsofworld.com.² Corporate income tax rate by country is sourced from the Organization for Economic Co-operation and Development (OECD). Additionally I collect data on bilateral investment treaties from United Nations Trade and Development Investment Policy Hub and double tax treaty data from OECD tax treaty database. For stock market data and foreign exchange rate data from 2000-2024, I use Compustat to collect: FX rates, stock market nominal returns, nominal exchange rates, real stock market returns, real exchange rate returns, and the monthly consumer price index. When calculating real exchange rate returns for countries in the European Union, I use the euro and the corresponding CPI for the EU. Thus all EU countries have the same exchange rate movements in the dataset. Data on the healthcare service industry is sourced from World Health Organization and World Development Indicators Databank. Lastly, countries serving as off-shore financial centers, such as Luxembourg, Cayman Islands, British Virgin Islands, where number of cross-border deals are disproportionate to population and GDP, are excluded.³

4.2 Variable Construction

The variables are grouped into five theoretical blocks: gravity (H_1), valuation arbitrage and deal-level controls (H_2), institutional and cultural distance (H_3), market depth and financing (H_4), and healthcare service industry and regulation (H_5).

4.2.1 Gravity

The gravity equation was originally introduced to international trade by Tinbergen (1962), extended by James E. Anderson in 1979 and again in 2003 by Anderson and Wincoop (2003). Following Anderson (1979) the gravity

²see Erel et al. (2012)p. 1053

³e.g. Cayman Islands record 55 cross-border deals, with a population of 74,457 in 2024 and in comparison Denmark had 43 at a population of 5.97 million and so on. Countries with missing trade data, total import and total export, from 2000-2024 were excluded from the dataset, countries like Mexico where data gaps appear more than once.

equation is estimated as:

$$M_{ijk} = \alpha_k Y_i^{\beta_k} Y_j^{\gamma_k} N_i^{\xi_k} N_j^{\varepsilon_k} d_{ij}^{\mu_k} U_{ijk} \quad (1)$$

where M_{ijk} is the dollar flow of good or factor k from country i to country j ; Y_i and Y_j are incomes in i and j ; N_i and N_j are populations in i and j ; d_{ij} is the distance between i and j ; and U_{ijk} is a lognormally distributed error term with $E(\ln U_{ijk}) = 0$. The coefficients $\beta_k, \gamma_k, \xi_k, \varepsilon_k, \mu_k$ are commodity-specific elasticities (Anderson 1979, p. 106). This equation model determines trade flow of commodities between countries. Countries with similar economic size, close geographic distance and similar population sizes, are more likely to attract foreign direct investments. For this thesis, I use the total value of GDP of $country_i$ and $country_j$ to measure economic size and transform to its logarithmic value. The population terms N_i and N_j in equation (1) are not estimated separately. Population and total GDP are highly correlated, and country population is mostly time-invariant over the sample period, so it is absorbed by the acquiror- and target-country fixed effects described in Section 5. Therefore, economic mass is captured monadically by GDP where bilateral trade is the dyadic linkage between two economies. The bilateral trade variable is constructed as:

$$Bilateral\ Trade_{ij,t} = Imports_{ij,t} + Exports_{ij,t} \quad (2)$$

Next, *Bilateral Trade* is converted to its logarithmic value:

$$\ln Bilateral\ Trade_{ij,t} = \ln(1 + Bilateral\ Trade_{ij,t}) \quad (3)$$

where $+1$ handles zero-valued observations. **Geographical distance** between i and j , is calculated using the great circle distance (Erel et al. 2012):

$$D = 6371.0 \times \arccos[\sin(\text{lat}_1) \cdot \sin(\text{lat}_2) + \cos(\text{lat}_1) \cdot \cos(\text{lat}_2) \cdot \cos(\text{lon}_2 - \text{lon}_1)]$$

Where (lat_1) is the latitude of the capital city in i and (lon_1) is the longitude of the capital city in i , (lat_2) is the latitude of j and (lon_2) is the longitude of the capital city in j . The trade data from UN Comtrade Database has inconsistent data measurements and data gaps which needs clarification. Each country reports its import and export, between itself and all countries in the world. Some observed values deviate significantly, e.g. i imports from j worth USD 100 million, i reports its import from j as USD 100 million and j reports its export to i as USD 20 million. In such cases I use the

primary value reported of the country that reports highest bilateral trade volume, i.e. $X_{ij} + M_{ij}$ denominated in USD. The following expression states its mathematical handling:

$$F_{ij,t} = \begin{cases} X_{ij,t}^i & \text{if } i \in \mathcal{R} \text{ and } X_{ij,t}^i \text{ reported} \\ M_{ij,t}^j & \text{else if } j \in \mathcal{R} \text{ and } M_{ij,t}^j \text{ reported} \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

Where $X_{ij,t}^i$ is the X_i/M_i reporting of i and $M_{ij,t}^j$ is the X_i/M_i reporting of j and $\mathcal{R}$ denotes the country whose report is retained, i.e. the one reporting the higher bilateral volume. If no trade value exists it defaults to zero and is excluded from the directed bilateral panel observations.⁴

4.2.2 Valuation Arbitrage and Deal-level controls

This block operationalises the valuation-arbitrage channel of hypothesis H_2 , together with the deal-level and firm-level controls used in the deal-size regressions. Under the valuation-arbitrage channel, bilateral differences in currency values, equity valuations, and the cost of capital generate financial incentives that direct cross-border deal flow. Consistent with Erel et al. (2012) I construct three country-pair variables that capture currency risk and exchange-rate movement, and an equity return differential that captures relative market valuation. Foreign exchange and stock market data are sourced from Compustat. **Foreign exchange volatility** captures the currency risk associated with a cross-border transaction between a country pair. I measure it as the annualised standard deviation of the monthly bilateral log exchange-rate return, computed over the twelve months of year t . Foreign Exchange rate 12 months volatility is estimated as:

$$\text{fx_volatility_12m_annual}_{ij,t} = \sqrt{12} \cdot \sqrt{\frac{1}{11} \sum_{m=1}^{12} (r_{ij,t,m} - \bar{r}_{ij,t})^2} \quad (5)$$

where $\bar{r}_{ij,t} = \frac{1}{12} \sum_{m=1}^{12} r_{ij,t,m}$ is the mean return. A more volatile bilateral exchange rate raises the uncertainty of the price and the financing of a foreign acquisition, so under the valuation-arbitrage channel higher volatility is expected to deter cross-border deal flow. **The annual exchange rate**

⁴e.g. Taiwan has no import and export data available in the UN Comtrade Database.

change captures the direction, rather than the dispersion, of the bilateral currency movement. It is the log change in the bilateral spot rate between the end of year $t - 1$ and the end of year t . The foreign exchange rate annual change between country pair is estimated as:

$$\text{fx_ln_change_annual}_{ij,t} = \ln\left(\frac{S_{ij,t}}{S_{ij,t-1}}\right) \quad (6)$$

$S_{ij,t}$ is the bilateral spot FX rate (units of currency j per unit of currency i) at the end of year t , or end of month m in year t and $r_{ij,t,m} = \ln(S_{ij,t,m}/S_{ij,t,m-1})$ is the monthly log return. The relative wealth channel of Froot and Stein (1991), claims a depreciation of the target currency relative to the acquiror currency lowers the effective price of target-country assets to a foreign acquiror, so the exchange-rate change is expected to be associated with greater inbound deal flow into the depreciating market. **The equity return differential** captures relative valuation across the two markets. It is the difference between the acquiror-country and target-country stock market returns, and a positive value indicates the acquiror market has outperformed the target market. Consistent with the valuation logic in Erel et al. (2012), acquirors in relatively highly valued markets are better placed to finance acquisitions of targets in relatively lower valued markets, so a positive differential is expected to raise outbound deal flow. I construct the differential in both nominal and real terms, where the real stock market return is adjusted for inflation of the two countries.

$$\text{Stock Market Return}_{ij} = R_{M,i,t} - R_{M,j,t} \quad (7)$$

A positive value indicates i has higher equity valuations than j . *The bilateral real exchange rate change* measures the currency movement in real terms, that is, after adjusting the nominal rate for the relative consumer price inflation of the two countries. The bilateral real exchange rate change is estimated as:

$$\Delta\text{RER}_{ij,t} = \ln\left(\frac{e_{ij,t}}{e_{ij,t-1}}\right) \times 100 \quad (8)$$

where the nominal change in equation (6) reflects the movement of the currency itself, the real change isolates the shift in relative purchasing power between the country pair, stripping out the part of the nominal movement that merely offsets an inflation differential. It enters the valuation-arbitrage channel through the same wealth mechanism as the nominal change. **Corporate**

income tax differential I construct *cit_signed_diff* as the signed differential between acquiror and target corporate income tax rates:

$$CIT_{ij,t} = CIT_{i,t} - CIT_{j,t} \quad (9)$$

Statutory CIT rates are sourced from the OECD Corporate Tax Rates database. A positive value indicates acquiror country has higher corporate taxes than the target, which under the tax-arbitrage channel predicts more outbound deal flow as firms seek to reduce effective tax burdens through foreign acquisition. **Bond yield differential** I measure the bilateral difference in the cost of capital as the differential between acquiror and target 10-year government bond yields,

$$\text{bond_yield_diff}_{ij,t} = B_{i,t} - B_{j,t} \quad (10)$$

A negative differential indicates the acquiror country faces a lower cost of capital than the target, which under the valuation-arbitrage channel predicts greater outbound deal flow, as acquirors with cheaper domestic financing are better placed to fund foreign acquisitions. Days from announcement to deal closure are recoded as missing for 445 observations (31% of deals) where the value is coded as 0, reflecting missing data in the source database rather than same-day closes. The remaining observations are winsorized at the 99th percentile to reduce the influence of a small number of deals with closing periods exceeding 1,000 days. Acquiror age is computed from the incorporation year of the legal acquiror entity as reported in Orbis M&A. For 17 acquirors (including ALJ Health Ltd, HCA, Laboratory Corporation of America Holdings, and Barclays PLC), the reported incorporation year reflected a corporate restructuring, holding-company formation, or subsidiary spin-out rather than the founding year of the underlying business. To be accurate I chose the founding year of the underlying business as the year of incorporation. Lastly, I winsorized firm variables *ebitda_margin_pre* and *leverage_pre* at the 1%/99% to reduce variance because of extreme outliers. I do not winsorize Log transformed variables, constructed variables, or where extreme outliers represent economic heterogeneity e.g. bilateral trade or domestic deal counts, these hold economic meaning. Argentina is excluded in Foreign Exchange variables due to hyperinflation in the sample period.

4.2.3 Institutional and Cultural Distance

Legal Distance contains three legal systems, civil law, common law, mixed or customary. Distance is measured as 0 if $country_i$ and $country_j$ both have e.g. civil law, 0.5 if i has civil law and j has common law, and 1 if i has civil law and j has mixed/other. Mixed includes if a country has both civil and common law, other includes customary law like islamic law. Legal distance is constructed as:

$$LD_{ij} = |L_i - L_j|, \quad L_c \in \{0, 0.5, 1\} \quad (11)$$

where L_i is the legal score of i and L_j is the legal score of j . **Religious Distance** following Stulz and Williamson (2003) I use four religious distributions, protestant, catholic, muslim, orthodox/other, the religious distance is then calculated as a composite religious distance of these four distributions, between i and j is constructed:

$$RD_{ij} = \sum_{r=1}^4 (s_{ri} - s_{rj})^2 \quad (12)$$

where s_{ri} is the population share of religion r in country i and s_{rj} is the population share of religion r in country j . **Cultural Distance** consists of Hofstede's four cultural dimensions presented in 1980, Power Distance Index (PDI), Individualism vs Collectivism (IDV), Masculinity vs Femininity (MAS), Uncertainty Avoidance Index (UAI) Hofstede (1980). Cultural distance between i and j is calculated as a composite of these four dimensions, following the Kogut and Singh (1988) approach:

$$CD_{ij} = \frac{1}{4} \sum_{k=1}^4 \frac{(I_{ki} - I_{kj})^2}{V_k} \quad (13)$$

where I_{ki} is the cultural score of i and I_{kj} is the cultural score of j on the cultural dimension k and V_k is the cultural variance.

4.2.4 Market Depth and Financing

This block proxies the depth of domestic M&A markets and the treaty environment that governs cross-border investment between a country pair. I measure domestic market depth by the count of domestic M&A deals completed within a country in a given year, constructed separately for the acquiror and the

target country. Following the treatment of count variables in the gravity block, the variable enters in logarithmic form as $\ln(1 + \text{domestic deal count})$, where the +1 accommodates country-years with zero recorded domestic deals. The acquiror-side measure, `ln_acq_domestic_deal_count`, proxies the depth and maturity of the acquiror’s home M&A market, since a deeper home market reflects greater financial development and accumulated ”dealmaking” capability, both of which lower the cost of foreign acquisition. The target-side measure, `ln_tgt_domestic_deal_count`, proxies the depth of the target market for acquisitions. The acquiror-side variable is a competing prediction, market-depth interpretation predicts a positive sign, since deeper home markets reflect dealmaking capacity, while a substitution interpretation predicts a negative sign, since an abundance of domestic targets reduces the incentive to acquire abroad. To capture the bilateral treaty environment, I construct two dummy variables. `BIT_in_force` equals one in country-pair-years where a bilateral investment treaty between *country-i* *country-j* is in force, and zero otherwise, since a bilateral investment treaty lowers expropriation risk and strengthens the legal protection of a foreign investment. `DTT_in_force` equals one where a double taxation treaty between the pair is in force, and zero otherwise, since a double taxation treaty eliminates the double taxation of cross-border income and reduces the effective tax cost of a foreign acquisition. Both treaties are coded as in force from their entry-into-force year onward, so the indicators vary within a country pair over time.

4.2.5 Healthcare service Industry and Regulation

To estimate healthcare variables effect on cross-border deal count, I use *healthcare expenditure as a percentage of total GDP* from World Development Indicators Databank as the primary healthcare variable. Healthcare expenditure absorbs the effect from aging populations, chronic diseases, hospital beds pr 1,000 inhabitant, technological progress, and healthcare human resources such as physicians and nurses. *Healthcare Regulation Openness Index* is a proxy based on the OECD FDI Regulatory Restrictiveness Index sourced from OECD Data Explorer. Lastly, I used the Product Market Regulation (PMR) index as an additional regulatory measurement. Variables like physicians per 1,000 inhabitants, universal healthcare index and percentage of the population older than 65, are used as robustness checks for healthcare variables since data availability is restricted (see Appendix D). In addition, I constructed dummy variables on regional specific regulatory frameworks, Food and Drug Adminis-

tration, European Medicines Agency and Other, I also added a UK Dummy that captures regulatory changes pre-, and-post Brexit. **Regulatory regime** I construct a four-tier classification of target-country pharmaceutical regulatory regimes. First, Tier 1 (FDA) contains the United States. Second, Tier 2 (EMA) contains the 24 EU/EEA member states and the United Kingdom before Brexit for the years 2000–2020. Tier 3 (Other Major) contains eight countries with established independent regulatory authorities: Switzerland, Japan, Canada, Australia, New Zealand, Singapore, Israel, and Hong Kong, plus the United Kingdom from 2021 onwards following the MHRA’s separation from EMA. Tier 4 (Other) contains the remaining countries in the sample and serves as the baseline category in regressions including `tgt_reg_tier` as a categorical regressor **Brexit difference-in-differences indicator**. The Brexit DiD variable `uk_post_brexit` equals one when the target country is the United Kingdom and the year is 2021 or later, and zero otherwise:

$$\text{UK_post_brexit}_{ijt} = \mathbb{1}[\text{tgt}_j = \text{GB}] \cdot \mathbb{1}[t \geq 2021] \quad (14)$$

This indicator captures the post-Brexit regulatory divergence following MHRA separation from EMA. Additionally I construct three interaction terms with acquiror-country regulatory tier `uk_post_x_acq_fda`, `uk_post_x_acq_ema`, `uk_post_x_acq_thermajor` to test if Brexit affected the inbound deal flow asymmetrically across regulatory backgrounds of the acquiror.

4.3 Summary Statistics

Table 2 reconstructs the deal matrix of Erel et al. (2012) on the healthcare-services sample, reporting the number of cross-border deals between country pairs. Domestic market depth is captured by the diagonal entries, which count domestic deals. Deal count cluster along the diagonal line and between developed economies, with the United States and bigger EU economies as dominant acquirors, the original matrix is shown in Appendix C. **Table 3** presents summary statistics for the country-pair-year panel used in the gravity regressions. The sample contains 54,144 directed bilateral observations spanning 47 countries from 2000-2024, with offshore financial centers excluded.⁵ The dependent variable `cb_deal_count` is highly zero-inflated: the mean is 0.029 deals per country-pair-year with a median of zero, while the maximum reaches 18 deals. The number of years where zero deals occurred between

⁵Bermuda, British Virgin Islands, Cayman Islands, Luxembourg.

The columns are acquiror countries labeled by ISO 2 codes. Destination/target countries are labeled on the left rows by country name and ISO2 code. Diagonal entries are the number of domestic deals. Non-diagonal entries are cross-border deals. The totals exclude domestic deals; thus, all totals are the number of cross-border deals for each country.

Destination	AD	AE	BE	BM	BR	VG	CA	KY	CL	CN	CO	CY	CZ	DK	EG	FR	DE	GR	HU	IN	IE	IT	JP	LU	MY	NZ	NO	PL	PT	RO	RU	SA	SG	SK	ZA	ES	CH	TR	AE	GB	US	Total				
Australia (AU)	792								2					2		1	1	1	1	1		1	2	1	1	1	10					8			2	1	1	6	18	849						
Austria (AT)	28															1	7						1	1																	1	39	1			
Belgium (BE)	2	73				1										11	1									3									1	2				3	98	19				
Bermuda (BM)			9	5	4													1																								5	318	1		
Brazil (BR)			301														3		13			1	1													1	2	1			1	83	1			
British Virgin Islands (VG)			7	41	12	8											3				1		1	1																	3	29	448	50		
Canada (CA)						408	1											4	1				2										2			4	1				2	3	35	1		
Cayman Islands (KY)			1	2		22														1																						2	9	667	41	
Chile (CL)																					1																						2	3	33	1
China (CN)	2	1	3	2	11	1,226											3	20					5	2									13								1	9	1,299	6		
Colombia (CO)	1					1	3									1	1		3																							2	168	1		
Cyprus (CY)																																											2	1	85	1
Czech Republic (CZ)	4																2	1			1												3			1	14	1			2	79	20			
Denmark (DK)	2																																										2	1	20	1
Egypt (EG)																																											2	9	441	1
Finland (FI)	1	1																																									2	9	441	1
France (FR)	16	3	5														400	1																									2	9	441	1
Germany (DE)																	2																										2	9	441	1
Greece (GR)	1	1	1														18	549																									2	9	441	1
Hong Kong (HK)																																											2	9	441	1
Hungary (HU)																																											2	9	441	1
India (IN)																																											2	9	441	1
Ireland (IE)	2																																										2	9	441	1
Israel (IL)																																											2	9	441	1
Italy (IT)	6																																										2	9	441	1
Japan (JP)																																											2	9	441	1
Luxembourg (LU)																																											2	9	441	1
Malaysia (MY)	1																																										2	9	441	1
Netherlands (NL)	4																																										2	9	441	1
New Zealand (NZ)	17																																										2	9	441	1
Norway (NO)																																											2	9	441	1
Poland (PL)	1																																										2	9	441	1
Portugal (PT)																																											2	9	441	1
Puerto Rico (PR)	1																																										2	9	441	1
Romania (RO)																																														

Source: Own Contribution; original matrix from Erel et al. (2012) is displayed in Appendix C.

country-pairs, establishes Pseudo-Poisson Maximum Likelihood estimation, as the correct estimator (Silva and Tenreyro 2006). acquiror- and target-side country-level variables share identical marginal distributions in the directed panel (e.g., `tgt_ln_total_gdp` and `acq_ln_total_gdp` both have a mean of 26.87, with a sd 1.34), reflecting that each country appears both as acquiror and target across the sample. Bilateral institutional and cultural distance variables show high cross-pair variation: legal distance ranges from 0 (identical legal traditions) to 1 (maximally distant), religious distance from 0 to 1.66, and cultural distance (Kogut-Singh, 4-dimension) from 0.04 to 9.84. Treaty coverage is modest, with bilateral investment treaties in force for 25% of pair-years and double-tax treaties for 9%. Approximately 16% of pair-years involve two EU members and 35% involve two OECD members.

Table 3: Summary Statistics: Country-Pair-Year Variables

	count	mean	sd	min	p50	max
cb_deal_count	58800	0.029	0.280	0.000	0.000	18.000
ln_total_gdp	58752	26.841	1.364	23.024	26.707	30.990
ln_distance_km	58800	8.454	1.009	4.935	8.863	9.896
ln_bilateral_trade	58336	17.654	8.184	0.000	20.749	28.060
domestic_deal_count	58800	14.494	43.008	0.000	2.000	580.000
ln_domestic_deal_count	58800	1.419	1.439	0.000	1.099	6.365
both_eu	58800	0.174	0.379	0.000	0.000	1.000
both_oecd	58800	0.363	0.481	0.000	0.000	1.000
BIT_in_force	58800	0.239	0.426	0.000	0.000	1.000
DTT_in_force	58800	0.095	0.294	0.000	0.000	1.000
legal_distance_bilateral	58800	0.389	0.372	0.000	0.500	1.000
religious_distance	58800	0.540	0.430	0.000	0.451	1.663
cultural_distance	54050	2.126	1.538	0.038	1.823	9.842
cultural_distance_6d	51750	1.999	1.183	0.045	1.848	7.642
bond_yield_diff	56448	0.000	3.173	-21.852	0.000	21.852
stock_return_diff_real	48906	0.000	1.545	-16.487	0.000	16.487
stock_return_diff_nom	48984	0.000	0.742	-16.552	0.000	16.552
fx_volatility_12m_annual	57984	3.192	36.543	0.000	0.027	1118.243
fx_ln_change_annual	58800	0.000	1.568	-25.716	0.000	25.716
cit_signed_diff	58254	0.000	10.699	-52.400	0.000	52.400
tgt_pmr_overall	45264	1.651	0.556	0.806	1.560	3.052
tgt_pmr_state_distortions	45264	1.695	0.614	0.824	1.621	3.206
tgt_pmr_entry_barriers	45264	1.606	0.536	0.768	1.525	3.071
tgt_sectpmr_pharmacy	44928	2.131	1.715	0.000	1.750	6.000
tgt_sectpmr_retail_distribution	44928	1.176	0.675	0.000	1.077	3.000
post_2021_pmr_vintage	58800	0.160	0.367	0.000	0.000	1.000
reg_ria	37536	0.433	0.170	0.100	0.423	0.824
reg_stakeholder	37536	0.405	0.181	0.089	0.396	0.833
reg_expost	37536	0.266	0.177	0.000	0.216	0.758
tgt_health_exp_pct_gdp	47136	8.084	2.596	1.842	8.111	18.516
tgt_hc_openness_index	56448	-0.031	0.433	-1.386	-0.047	1.319
ln_hc_svc_revenue_ratio	27065	0.178	3.318	-6.908	0.073	15.138
Observations	54144					

Sample: bilateral country-pair-year observations, 2000-2024, excluding offshore financial centers (VG, KY, BM, LU). Variables ordered by theoretical block: outcome, gravity, institutional, cultural, FX/macro, healthcare.

Financial market variables display the symmetric distributions around zero: the bond yield differential ranges from -21.9 to $+21.9$ percentage points, and the FX log change is approximately symmetric with a wide tail driven by emerging-market volatility events. The corporate income tax differential (`cit_signed_diff`) ranges from -52.4 to $+52.4$ percentage points, reflecting substantial sovereign tax-policy heterogeneity. Coverage gap appear for select variables: stock return differentials are missing for approximately 16% of observations due to Compustat coverage, and `ln_hc_svc_revenue_ratio` is restricted to 26,917 observations due to the ratio's definition (both countries must have positive healthcare service revenue). **Table 4** reports summary statistics for the deal-level sample used in firm and deal-level regressions

(Blocks 2–3). The sample contains 1,463 completed cross-border healthcare M&A transactions with sufficient disclosure for inclusion in deal-level regressions. The transactions are predominantly majority-stake acquisitions (95%) with mean final stake of 95%, and 78% are full acquisitions. Only 28% involve same-NAICS4 industry classification, indicating that cross-border healthcare M&A is largely diversifying or vertical rather than horizontal integration. The mean acquiror is 21 years old at deal announcement, with 25% classified as serial acquirors (three or more prior transactions). Firm-fundamental variables show smaller sample sizes (N=540–1,463) due to Compustat and Orbis coverage restrictions on acquiror financials, particularly for non-US listed acquirors and private firms, deal-level regressions are estimated on the intersection of the deal sample and the firm-fundamentals coverage.

Table 4: Summary Statistics: Deal-Level Variables

	count	mean	sd	min	p50	max
deal_size_log	540	10.013	2.358	1.374	10.099	16.194
is_majority	1370	0.953	0.213	0.000	1.000	1.000
is_full	1370	0.785	0.411	0.000	1.000	1.000
same_naics4	1458	0.281	0.450	0.000	0.000	1.000
days_to_close	1463	32.541	154.492	0.000	0.000	2671.000
final_stake_pct	1348	95.090	13.116	50.000	100.000	100.000
prior_total_deals	1463	2.668	5.887	0.000	0.000	72.000
serial_acquirer	1463	0.244	0.430	0.000	0.000	1.000
acq_age_at_deal	1252	21.627	25.982	0.000	14.000	222.000
Log of (acquirer age at deal + 1)	1252	2.606	1.055	0.000	2.708	5.407
Acq_LogTA	1056	12.510	2.873	0.238	12.494	21.364
ebitda_margin_pre	604	-0.579	3.588	-28.382	0.123	0.780
leverage_pre	848	0.345	0.740	0.000	0.182	6.071
Observations	1463					

Sample: completed cross-border healthcare deals with disclosed deal value, 2000-2024, offshore-excluded. Variables ordered by theoretical block: outcome, deal characteristics, acquirer experience, firm fundamentals. Firm fundamentals sample is smaller due to Orbis coverage restrictions.

5 Methodology

5.1 Empirical Strategy

This section covers the empirical strategy I use to test hypotheses $H_1 - H_5$. The empirical strategy covers three steps, first the estimation of the baseline gravity specification on the country-pair-year panel (Block 1) using Pseudo-Poisson

Maximum Likelihood (PPML) by Silva and Tenreyro (2006). The baseline regression is structured as a cascade, gradually adding each theoretical block, to decompose the contribution of each hypotheses. Second, I estimate deal-level and firm-level regressions on their respective subsamples of cross-border healthcare deals to test deal-characteristics and acquiror-firm determinants. Third, I estimate the robustness specifications (**Blocks 9, 11, 12, 13**) testing the headline findings using alternative specifications, sample-splits and identification strategies. All regressions apply three-way fixed effects (year, acquiror-country, target-country) with standard errors clustered at the country-pair level.

5.2 Baseline regression

The dependent variable in the headline specification is the count of completed cross-border healthcare M&A transactions from acquiror country i to target country j in year t , denoted as $D_{ij,t}$. Because the dependent variable is a non-negative number with approximately 97.6% of observations equalling zero, gravity equation cannot be estimated directly via OLS. Following Silva and Tenreyro (2006), I estimate the gravity model via Pseudo-Poisson Maximum Likelihood (PPML), which naturally handles country-pair-years and years where zero deals occurred. OLS would require dropping all zero-count observations, and keeping only outcome variables, that is country-pair-years where deals happened, since gravity variables are logarithmic values. PPML preserves the gravity structure and produces consistent estimates under heteroskedasticity (Silva and Tenreyro 2006). The baseline specification can thus be estimated:

$$\mathbb{E}[D_{ij,t} \mid \mathbf{X}_{ij,t}] = \exp(\alpha + \mathbf{X}_{ij,t}'\boldsymbol{\beta} + \lambda_t + \lambda_i^{acq} + \lambda_j^{tgt}) \quad (15)$$

where $\mathbf{X}_{ij,t}$ is the vector of bilateral and country-specific regressors described in Section 5, λ_t are year fixed effects, λ_i^{acq} are acquiror-country fixed effects, and λ_j^{tgt} are target-country fixed effects. The full specification, separating

regressors by hypothesis block:

$$\begin{aligned} \mathbb{E}[D_{ij,t} \mid \mathbf{X}_{ij,t}] = \exp & \left(\lambda_t + \lambda_i^{acq} + \lambda_j^{tgt} + \alpha \right. \\ & + \beta_1 \ln GDP_{i,t} + \beta_2 \ln GDP_{j,t} + \beta_3 \ln Dist_{ij} \\ & + \beta_4 \ln Trade_{ij,t} + \beta_5 BothEU_{ij} + \beta_6 BothOECD_{ij} \\ & + \beta_7 BondYieldDiff_{ij,t} + \beta_8 \sigma_{ij,t}^{FX} + \beta_9 \Delta \ln FX_{ij,t} \\ & + \beta_{10} CITDiff_{ij,t} + \beta_{11} LegalDist_{ij} + \beta_{12} RelDist_{ij} \\ & + \beta_{13} CultDist_{ij} + \beta_{14} \ln DomMA_{i,t}^{acq} + \beta_{15} \ln DomMA_{j,t}^{tgt} \\ & + \beta_{16} BIT_{ij,t} + \beta_{17} DTT_{ij,t} + \beta_{18} HealthExp_{j,t} \\ & \left. + \beta_{19} PMR_{j,t} + \beta_{20} HCOpenness_{j,t} + \beta_{21} \ln HCRevRatio_{ij,t} \right) \end{aligned} \quad (16)$$

5.3 Identification

Variables that causes friction in cross-border M&A deals are (distance, trade, legal/religious/cultural distance, treaties) are identified from cross-pair variation, since these are time-invariant or slowly-changing bilateral characteristics. The distance variables are constant for each country-pair and their respective coefficients represents bilateral frictions, which reduces deal count. Identification of time-varying country-pair variables (FX channels, bond yield differential, CIT differential, bilateral trade) comes from within-country-pair temporal variation. The three-way fixed effects structure absorbs year-specific shocks (λ_t) like GFC or Covid, time-invariant acquiror characteristics (λ_i^{acq}), and time invariant target characteristics (λ_j^{tgt}), these coefficients absorbs deviations from these averages. Identification of target-country regulatory and healthcare variables (PMR, healthcare expenditure, healthcare regulatory openness, regulatory regime tier) comes from within-country time variation. Because acquiror and target fixed effects absorb time-invariant country level effects, identification of these variables requires higher within-country variation within the sample period 2000-2024. Block 9 explicitly tests if *tgt_pmr_overall* identifies off within-country or cross-country variation by varying the fixed effects structure, across the year, acquiror country and target country and four additional fixed effects structures. Lastly, I do not claim a causal relationship between the regressed variables and the estimated coefficients. The specification identifies correlations between bilateral M&A deal flows and

country-pair characteristics, following the empirical approach of Head and Mayer (2014).

5.4 Block Compositions

Block 1: Gravity Determinants Block 1 is the regression specification that follows a cascade structure across five hypothesis blocks $H_1 - H_5$, each motivated by a theory in Section 3 and operationalized in Section 4. Block 1 (Table B1) presents the cumulative cascade adding hypothesis blocks gradually, each column is described with its results in section 6.1. **Block 2: Deal Determinants** estimates the deal-level variables on the sample of 1,463 cross-border healthcare services deals with disclosed deal values. The dependent variable is the logarithmic value of deal values (*deal_size_log*) estimated using *reghdfe* with deal year, acquiror-country, and target-country fixed effects and standard errors clustered at the country-pair level. Regressors of deal specific characteristic variables includes majority-acquisition (*is_majority*), full acquisition (*is_full*), same industry code (*same_naics4*), days to close (*days_to_close*), and final stake percentage (*final_stake_pct*). **Block 3: Firm Determinants** estimates the firm-level variables on the sample of 1,463 cross-border healthcare services deals. Similar as deal determinants, the dependent variable is the logarithmic value of deal values (*deal_size_log*) estimated as *reghdfe* with deal year, acquiror-country, and target-country fixed effects and standard errors clustered at the country-pair level. Regressors of firm specific characteristic variables include: logarithmic value of acquirors total assets (*Acq_LogTA*), pre-deal EBITDA margin (*ebitda_margin_pre*), acquiror firm pre-deal leverage (*leverage_pre*), acquiror total count of prior deals (*prior_total_deals*), acquiror is a serial acquiror (*serial_acquiror*, ≥ 3 deals), and the logarithmic value of acquirors age at time of deal (*ln_acq_age_at_deal*). **Block 4: Healthcare Industry** estimates the industry-level variables on the sample of 15,097 country-pair-year observations. The dependent variable is deal count estimated using *ppmlhdfe* with deal year, acquiror country and target country fixed effects and standard errors clustered at the country-pair level. Industry specific regressors include: target country overall product market regulation index score (*tgt_pmr_overall*), target country healthcare expenditure as a percentage of total GDP (*tgt_health_exp_pct_gdp*), target country healthcare openness index score (*tgt_hc_openness_index*), logarithmic value of healthcare service firms *acquiror/target* ratio (*ln_hc_svc_revenue_ratio*).

5.5 Robustness panels

Beyond the headline cascade of regressions, I estimate five robustness panels to test the findings of $H_1 - H_5$ under alternative specifications, different sample splits, and identifications (Appendix D). Block 5 is the equity channel (real vs nominal stock return differential) and tests pre-/post-global financial crisis sample stability. Block 6 compares $\ln(1+\text{count})$ versus $\ln(\text{value})$ specifications of domestic M&A activity. Block 7 tests cultural distance robustness using Hofstede’s 6-dimension update. Block 8 substitutes a healthcare-specific four-tier regulatory regime variable and adds Brexit DiD analysis specification to test the asymmetric effects of post-2021 regulatory divergence by acquiror regulatory background. Block 9 varies the fixed effects structure (Year only, Year + Acq only, 3-way FE) to test if the headline *tgt_pmr_overall* result identifies on within-country or cross-country variation. Block 10 tests country-level institutional variables (Disclosure Index, World Governance Indicators Composite) under multiple FE specifications. Block 11 extends the healthcare characterization with healthcare *demand-side* variables like percentage of population 65+, universal healthcare coverage (UHC), and *supply-side* healthcare variables like physicians pr 1,000 citizens. Block 12 tests the OECD FDI Regulatory Restrictiveness Index alongside the headline regulatory measures. Block 13 decomposes the Product Market Regulation variable by subcomponents; state distortions and entry barriers.

6 Results

6.1 Main Results - Headline Specification

Table 5 presents the headline PPML cascade. Column (1) reports the gravity baseline regression, with GDP measures, geographical distance in kilometers, bilateral trade, and the H_4 domestic market-depth ($\text{count}_{ij}^{\text{dom}}$) and treaty controls (*both eu, both oecd*). Column (2) adds the H_3 institutional variables (bilateral investment treaty, double tax treaty, legal distance). Column (3) adds religious and cultural distance. Column (4) adds the H_2 valuation arbitrage variables (*FX volatility 12 months annual, FX ln change annual, Corporate Income Tax Difference*). Column (5) adds the H_5 healthcare-industry (*target country healthcare expenditure as a percentage of GDP, ln revenue ratio (acquiror/target)*) and regulation variables (*target country product market regulation, healthcare openness index*). The cumulative pseudo-

R^2 rises from 0.439 in the gravity baseline to 0.500 in the full specification. Because the sample composition changes across the cascade, the marginal contribution of each block is assessed separately on a common sample in Section 7.6.

Table 5: B1: Country-Pair-Year Cascade (PPML, Year + Acq + Tgt FE)

	(1) Gravity	(2) +Institutional	(3) +Cultural	(4) +FX/Macro	(5) +Healthcare
tgt_ln_total_gdp	0.560* (0.294)	0.560** (0.280)	0.484* (0.281)	0.481* (0.270)	0.404 (0.308)
acq_ln_total_gdp	-0.383 (0.273)	-0.326 (0.253)	-0.299 (0.252)	-0.295 (0.249)	-0.421 (0.362)
ln_distance_km	-0.721*** (0.230)	-0.747*** (0.097)	-0.705*** (0.068)	-0.704*** (0.069)	-0.298** (0.138)
ln_bilateral_trade	0.398* (0.230)	0.281*** (0.076)	0.216*** (0.041)	0.217*** (0.041)	0.604*** (0.119)
Tgt domestic deal count	0.025 (0.065)	0.022 (0.064)	0.021 (0.065)	0.016 (0.063)	0.046 (0.079)
Acq domestic deal count	0.276*** (0.058)	0.269*** (0.058)	0.251*** (0.058)	0.257*** (0.058)	0.235*** (0.085)
both_eu	-0.514** (0.207)	-0.666*** (0.189)	-0.600*** (0.182)	-0.606*** (0.183)	-0.419* (0.250)
both_oecd	1.187*** (0.345)	0.994*** (0.329)	0.419 (0.268)	0.450* (0.269)	1.201*** (0.292)
BIT_in_force		-0.162 (0.215)	0.024 (0.189)	0.028 (0.189)	-0.234 (0.198)
DTT_in_force		0.515*** (0.199)	0.602*** (0.188)	0.595*** (0.194)	0.808*** (0.228)
legal_distance_bilateral		-0.978*** (0.150)	-0.777*** (0.149)	-0.786*** (0.148)	-0.705*** (0.175)
religious_distance			-1.287*** (0.210)	-1.289*** (0.210)	-1.518*** (0.383)
cultural_distance			-0.250*** (0.053)	-0.248*** (0.053)	-0.153* (0.089)
10Y bond yield differential				-0.000 (0.034)	-0.020 (0.048)
fx_volatility_12m_annual				0.139 (0.138)	0.071 (0.202)
fx_ln_change_annual				0.021 (0.031)	0.018 (0.039)
cit_signed_diff				0.016*** (0.006)	0.026*** (0.007)
tgt_health_exp_pct_gdp					0.073 (0.107)
tgt_pmr_overall					2.172** (1.002)
tgt_hc_openness_index					0.522 (0.621)
Revenue ratio					-0.061 (0.045)
Constant	-12.203 (10.152)	-10.254 (9.694)	-6.653 (9.590)	-6.727 (9.558)	-18.604 (12.765)
Observations	49704	49704	45432	45432	15097
Pseudo R-sq	0.439	0.452	0.466	0.467	0.500
Wald chi2	410.12	468.01	880.78	931.91	625.48
Prob χ^2	0.000	0.000	0.000	0.000	0.000

Standard errors in parentheses. Three-way FE: Year + acq_iso2 + tgt_iso2. Pair clustering (Erel, Liao & Weisbach 2012). Bond yields entered as differential (acq - tgt); stock returns as real differential. Healthcare service revenue ratio in logs. PPML on cross-border healthcare deal count. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Gravity (H_1) The three H_1 gravity variables enter in column (1) and carry through the full cascade. Target-country GDP enters positive and marginally significant at $\beta = 0.560$ ($p < 0.10$) in column (1), holds at $\beta = 0.560$ ($p < 0.05$) in column (2), moderates to $\beta = 0.484$ ($p < 0.10$) in column (3) and $\beta = 0.481$ ($p < 0.10$) in column (4), and loses significance in the full specification at $\beta = 0.404$ ($p > 0.10$). acquiror-country GDP is statistically null across all five columns, entering at $\beta = -0.383$ in column (1) and remaining insignificant through $\beta = -0.421$ in column (5). The distance elasticity is negative and significant throughout, entering at $\beta = -0.721$ ($p < 0.01$) in column (1), holding between -0.747 and -0.704 across columns (2) to (4), and reduces to $\beta = -0.298$ ($p < 0.05$) in the full specification. Thus a one-percent increase in bilateral distance is associated with a 0.298-percent decrease in expected deal count. Bilateral trade is positive and significant in every column, entering at $\beta = 0.398$ ($p < 0.10$) in column (1), settling in the 0.216–0.281 range across columns (2) to (4), and strengthening to $\beta = 0.604$ ($p < 0.01$) in the full specification. The distance and trade coefficients carry the signs predicted by H_1 in every column, acquiror GDP does not, and target GDP is significant only outside the full specification. **Valuation arbitrage (H_2)** The four H_2 variables enter the cascade in column (4) and carry through to the full specification in column (5). The bilateral bond yield differential is statistically null in both columns, entering at $\beta = -0.000$ ($p > 0.10$) in column (4) and $\beta = -0.020$ ($p > 0.10$) in column (5), the point estimate does not approach its standard error in either specification. FX volatility is likewise null, with $\beta = 0.139$ in column (4) and $\beta = 0.071$ in column (5), neither significant at conventional thresholds. The FX log change is null throughout, entering at $\beta = 0.021$ in column (4) and $\beta = 0.018$ in column (5). The corporate income tax differential is the only H_2 variable that produces a coefficient: it enters positive and significant at $\beta = 0.016$ ($p < 0.01$) in column (4) and strengthens to $\beta = 0.026$ ($p < 0.01$) in the full specification. A one-percentage-point increase in the acquiror-minus-target CIT differential is associated with a 2.6-percent increase in expected cross-border deal count. Of the four valuation-arbitrage channels, only tax arbitrage is supported under H_2 , the bond yield and foreign exchange channels are not. **Institutional and cultural distance (H_3)** The three institutional and cultural distance variables enter at different stages of the cascade. Legal distance enters in column (2) at $\beta = -0.978$ ($p < 0.01$) and remains negative and significant across all subsequent columns, moderating to $\beta = -0.777$ in column (3), $\beta = -0.786$ in column (4), and $\beta = -0.705$ ($p < 0.01$) in the full specification. Religious distance enters in

the cultural block at $\beta = -1.287$ ($p < 0.01$), is stable through the FX/macro block at $\beta = -1.289$, and strengthens to $\beta = -1.518$ ($p < 0.01$) in the full healthcare specification. Cultural distance, measured as the four-dimension Kogut-Singh composite, enters in column (3) at $\beta = -0.250$ ($p < 0.01$), holds at $\beta = -0.248$ in column (4), and attenuates to $\beta = -0.153$ ($p < 0.10$) in column (5). All three variables carry the negative sign predicted by H_3 in every column in which they appear. Religious distance is the largest in magnitude, followed by legal distance, with cultural distance the smallest, the ordering is stable across the cascade. **Market depth and financing (H_4)** The H_4 variables enter the cascade at two stages: the domestic deal counts and the both-EU and both-OECD indicators in column (1), and the bilateral treaty indicators in column (2). The acquiror’s log domestic deal count is positive and significant across all five columns, entering at $\beta = 0.276$ ($p < 0.01$) in column (1) and holding in the 0.235–0.276 range through to $\beta = 0.235$ ($p < 0.01$) in the full specification. The target’s log domestic deal count is statistically null throughout, entering at $\beta = 0.025$ in column (1) and remaining insignificant through $\beta = 0.046$ in column (5). The DTT indicator enters in column (2) at $\beta = 0.515$ ($p < 0.01$) and is positive and significant across all subsequent columns, strengthening to $\beta = 0.808$ ($p < 0.01$) in the full specification. The BIT indicator enters in column (2) at $\beta = -0.162$ and is statistically null in every column in which it appears. Both-OECD is positive throughout, entering at $\beta = 1.187$ ($p < 0.01$) in column (1), holding at $\beta = 0.994$ ($p < 0.01$) in column (2), losing significance in the cultural block at $\beta = 0.419$ ($p > 0.10$), recovering to $\beta = 0.450$ ($p < 0.10$) in column (4), and reaching $\beta = 1.201$ ($p < 0.01$) in the full specification. Both-EU is negative and significant in every column, entering at $\beta = -0.514$ ($p < 0.05$) in column (1), strengthening to the -0.600 to -0.666 range across columns (2) to (4), and moderating to $\beta = -0.419$ ($p < 0.10$) in the full specification. Of the H_4 variables, the acquiror domestic deal count, DTT, and both-OECD carry positive and significant coefficients consistent with the market-depth and financing prediction, the target domestic deal count and BIT are null, and both-EU enters with a negative sign throughout. **Regulation (H_5)** The four H_5 healthcare variables enter only in the full specification in column (5). Target-country product market regulation is positive and significant at $\beta = 2.172$ ($p < 0.05$), the largest coefficient in the headline specification. Target-country health expenditure as a share of GDP is statistically null at $\beta = 0.073$ ($p > 0.10$). The healthcare regulatory openness index is positive but statistically null at $\beta = 0.522$ ($p > 0.10$), with a point estimate below its

standard error. The log healthcare service revenue ratio between acquiror and target is negative and statistically null at $\beta = -0.061$ ($p > 0.10$), the negative point estimate is directionally consistent with deal flow toward relatively larger target-side healthcare markets but does not reach conventional significance. Among the four H_5 variables, only product market regulation produces an effect on cross-border deal count.

6.2 Deal-level results

Table 6 reports the deal-level cascade, estimated by OLS on log deal value for the sample of cross-border healthcare transactions with disclosed deal value (N falls from 441 to 348 across the cascade as acquiror-experience variables are added, adjusted $R^2 = 0.357$ in the full specification). The gravity coefficients reverse sign relative to the deal-count specification in Table 5. I find bilateral distance positive and significant at $\beta = 1.375$ ($p < 0.01$) and bilateral trade positive and significant at $\beta = 1.229$ ($p < 0.01$), where both variables were negative or smaller in the deal-count model. Religious distance is likewise positive and significant at $\beta = 3.980$ ($p < 0.01$), and common language is marginally negative at $\beta = -1.931$ ($p < 0.10$), legal distance and cultural distance are insignificant. Common legal origin enters positive and marginally significant in the cultural block at $\beta = 1.246$ ($p < 0.10$), strengthens to $\beta = 2.939$ ($p < 0.05$) when deal characteristics are added, and loses significance in the full specification at $\beta = 2.242$. Among the deal characteristics, I find final stake percentage positive and significant at $\beta = 0.036$ ($p < 0.05$) and days-to-close positive and significant at $\beta = 0.006$ ($p < 0.05$), while the full-acquisition indicator is negative at $\beta = -0.935$ ($p < 0.10$), majority-acquisition status and same-NAICS4 industry classification are insignificant. The bilateral investment treaty indicator is positive across the cascade ($\beta = 1.439$, $p < 0.10$ in the full specification). The acquiror-experience variables, prior deal count, serial-acquiror status, and acquiror age, are all insignificant.

Table 6: B2: Deal-Level Cascade (reghdfe on deal_size_log)

	(1) Gravity	(2) +Institutional	(3) +Cultural	(4) +Deal Chars	(5) +Acq Experience
ln_distance_km	0.784** (0.301)	0.885*** (0.325)	1.055*** (0.374)	1.084*** (0.408)	1.375*** (0.419)
ln_bilateral_trade	0.526** (0.243)	0.594** (0.269)	0.817** (0.329)	0.895** (0.404)	1.229*** (0.436)
both_eu		-0.110 (0.547)	-0.157 (0.572)	-0.379 (0.529)	-0.618 (0.575)
both_oecd		2.028*** (0.776)	1.573 (1.095)	1.137 (1.425)	1.299 (1.645)
BIT_in_force		0.981** (0.447)	1.038** (0.510)	1.752*** (0.617)	1.439* (0.788)
DTT_in_force		0.188 (0.538)	0.220 (0.582)	0.619 (0.749)	-0.245 (0.671)
legal_distance_bilateral		0.059 (0.313)	0.689 (0.785)	2.314 (1.581)	1.494 (1.942)
religious_distance			1.233 (0.903)	3.567*** (1.141)	3.980*** (1.088)
cultural_distance			0.179 (0.214)	-0.078 (0.288)	0.087 (0.324)
common_language			-1.117 (0.906)	-1.689 (1.073)	-1.931* (1.081)
common_legal_origin			1.246* (0.663)	2.939** (1.198)	2.242 (1.610)
is_majority				1.924** (0.788)	1.096 (0.797)
is_full				-1.140** (0.524)	-0.935* (0.507)
same_naics4				0.234 (0.268)	-0.173 (0.251)
days_to_close				0.001 (0.002)	0.006** (0.002)
final_stake_pct				0.039** (0.015)	0.036** (0.015)
prior_total_deals					0.053 (0.038)
serial_acquiror					0.328 (0.447)
Acquiror Age					0.139 (0.136)
Constant	-8.811 (8.010)	-12.745 (8.982)	-20.027* (10.689)	-28.004** (12.918)	-37.975*** (13.899)
Observations	441	441	431	387	348
R-sq	0.402	0.413	0.422	0.469	0.522
Adj R-sq	0.255	0.258	0.262	0.302	0.357
F-stat	3.39	2.62	2.02	2.83	3.84
Prob > F	0.036	0.014	0.031	0.001	0.000

Standard errors in parentheses. Pair clustering (AcqCountry x TgtCountry); FE: DealYear + AcqCountry + TgtCountry (additive 3-way). Sample: completed cross-border healthcare deals, offshore-excluded, conditional on disclosed deal value. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.3 Firm-level results

Table 7 extends the deal-level specification with acquiror firm fundamentals, which restricts the sample to 175 transactions with non-missing Compustat and Orbis acquiror financials. Adjusted R^2 rises from 0.357 to 0.673 across

the cascade, with the addition of acquiror size producing the largest single gain: the unadjusted R^2 rises from 0.522 to 0.629 when log total assets enters in column 2 and to 0.815 in column 3. I find the log of acquiror total assets positive and significant at $\beta = 0.520$ ($p < 0.01$) in the full specification, indicating that larger acquirors undertake larger transactions. The pre-deal EBITDA margin is negative and significant at $\beta = -0.122$ ($p < 0.01$), while pre-deal leverage, prior deal count, serial-acquiror status, acquiror age, and the interaction between acquiror size and prior deal count are all insignificant. The gravity and cultural variables that were significant in the deal-level specification lose significance once firm fundamentals enter: bilateral distance, bilateral trade, and religious distance are all insignificant in columns 2 to 5, indicating that acquiror size absorbs the explanatory content these variables carried in Table 6. Among the deal characteristics, majority-acquisition status strengthens from insignificant in column 1 to $\beta = 3.448$ ($p < 0.01$) in the full specification, and final stake percentage strengthens to $\beta = 0.054$ ($p < 0.01$), the full-acquisition indicator remains negative and significant at $\beta = -1.437$ ($p < 0.05$). Both-OECD becomes positive and significant once firm fundamentals enter ($\beta = 5.573$, $p < 0.01$), while common legal origin and the double tax treaty indicator both turn negative and marginally significant in the restricted sample ($\beta = -3.134$ and $\beta = -2.618$, $p < 0.10$). The bilateral investment treaty indicator is collinear in the restricted sample and is dropped from columns 3 to 5.

Table 7: B3: Firm-Level Cascade (reghdfe on deal_size_log + firm fundamentals)

	(1) Deal Base	(2) +Size	(3) +Profitability	(4) +Leverage	(5) +Size×Exp
ln_distance.km	1.375*** (0.419)	0.327 (0.448)	0.327 (0.597)	0.326 (0.599)	0.358 (0.635)
ln_bilateral_trade	1.229*** (0.436)	0.199 (0.446)	0.503 (0.539)	0.503 (0.542)	0.517 (0.559)
both_eu	-0.618 (0.575)	-0.635 (0.607)	-0.426 (0.819)	-0.433 (0.830)	-0.313 (0.966)
both_oecd	1.299 (1.645)	1.566 (1.454)	5.360*** (1.771)	5.481*** (1.600)	5.573*** (1.615)
BIT_in_force	1.439* (0.788)	1.221 (0.859)	0.000 (.)	0.000 (.)	0.000 (.)
DTT_in_force	-0.245 (0.671)	0.057 (0.933)	-2.518** (1.252)	-2.541* (1.310)	-2.618* (1.357)
legal_distance_bilateral	1.494 (1.942)	1.108 (1.866)	-3.079 (1.859)	-3.064 (1.939)	-2.947 (1.911)
religious_distance	3.980*** (1.088)	2.880*** (1.098)	0.737 (1.853)	0.748 (1.855)	0.645 (1.724)
cultural_distance	0.087 (0.324)	-0.219 (0.301)	0.104 (0.312)	0.105 (0.318)	0.080 (0.290)
common_language	-1.931* (1.081)	0.554 (1.294)	-0.050 (1.267)	-0.041 (1.302)	0.121 (1.189)
common_legal_origin	2.242 (1.610)	1.009 (1.510)	-3.126* (1.617)	-3.112* (1.695)	-3.134* (1.698)
is_majority	1.096 (0.797)	1.430* (0.756)	3.476*** (0.737)	3.469*** (0.764)	3.448*** (0.794)
is_full	-0.935* (0.507)	-0.684 (0.473)	-1.421** (0.667)	-1.431* (0.722)	-1.437** (0.718)
same_naics4	-0.173 (0.251)	-0.035 (0.296)	-0.339 (0.390)	-0.336 (0.408)	-0.347 (0.418)
days_to_close	0.006** (0.002)	0.004* (0.002)	0.001 (0.002)	0.001 (0.002)	0.001 (0.002)
final_stake_pct	0.036** (0.015)	0.036*** (0.012)	0.054*** (0.016)	0.054*** (0.019)	0.054*** (0.019)
prior_total_deals	0.053 (0.038)	0.002 (0.040)	0.032 (0.021)	0.033 (0.021)	-0.131 (0.359)
serial_acquirer	0.328 (0.447)	0.306 (0.345)	-0.010 (0.256)	-0.015 (0.266)	0.107 (0.397)
Acquiror Age	0.139 (0.136)	-0.146 (0.115)	-0.016 (0.188)	-0.016 (0.187)	-0.025 (0.189)
Acq_TA		0.371*** (0.053)	0.532*** (0.068)	0.533*** (0.069)	0.520*** (0.079)
ebitda_margin_pre			-0.118*** (0.047)	-0.118** (0.047)	-0.122*** (0.043)
leverage_pre				0.027 (0.281)	-0.003 (0.282)
TA_x_priordeals					0.010 (0.022)
Constant	-37.975*** (13.899)	-8.818 (14.487)	-20.919 (17.170)	-21.081 (17.607)	-21.572 (18.066)
Observations	348	322	175	175	175
R-sq	0.522	0.629	0.815	0.815	0.816
Adj R-sq	0.357	0.487	0.679	0.676	0.673
F-stat	3.84	6.99	24.99	23.58	24.19
Prob > F	0.000	0.000	0.000	0.000	0.000

Standard errors in parentheses. Acquirer firm fundamentals progressively added; sample restricted to disclosed-value deals with non-missing acquirer Orbis fundamentals. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.4 Healthcare channel decomposition

Table 8 decomposes the healthcare block by adding the four healthcare variables sequentially to the headline specification. Product market regulation enters first (column 1) and is positive and significant throughout, at $\beta = 2.106$ ($p < 0.01$) in column 1 and $\beta = 2.172$ ($p < 0.05$) in the full specification. Target health expenditure as a share of GDP enters in column 2 and is insignificant in every column in which it appears ($\beta = 0.073$ in the full specification). The healthcare regulatory openness index enters in column 3 and is positive but insignificant ($\beta = 0.523$ in column 3, $\beta = 0.522$ in the full specification). The log healthcare service revenue ratio enters in the full specification and is negative and insignificant at $\beta = -0.061$. Of the four healthcare variables, only product market regulation produces an effect on cross-border deal count. The sample falls from approximately 34,000 observations in columns 1 to 3 to 15,097 in the full specification, as the healthcare service revenue ratio requires positive healthcare service revenue in both countries, the pseudo- R^2 rises from 0.473 to 0.500 across this transition.

Table 8: B4: Healthcare Channel Decomposition (PPML, Year + Acq + Tgt FE)

	(1) +Reg	(2) +Spend	(3) +FDI	(4) Full
tgt_ln_total_gdp	0.564* (0.289)	0.568* (0.291)	0.645** (0.296)	0.404 (0.308)
acq_ln_total_gdp	-0.337 (0.296)	-0.342 (0.296)	-0.342 (0.297)	-0.421 (0.362)
ln_distance_km	-0.618*** (0.087)	-0.624*** (0.086)	-0.624*** (0.086)	-0.298** (0.138)
ln_bilateral_trade	0.274*** (0.062)	0.269*** (0.060)	0.269*** (0.060)	0.604*** (0.119)
Tgt domestic deal count	0.052 (0.067)	0.063 (0.067)	0.038 (0.065)	0.046 (0.079)
Acq domestic deal count	0.240*** (0.065)	0.244*** (0.065)	0.247*** (0.067)	0.235*** (0.085)
both_eu	-0.571*** (0.193)	-0.579*** (0.193)	-0.574*** (0.194)	-0.419* (0.250)
both_oecd	0.847*** (0.271)	0.849*** (0.271)	0.829*** (0.272)	1.201*** (0.292)
BIT_in_force	-0.242 (0.173)	-0.231 (0.171)	-0.244 (0.172)	-0.234 (0.198)
DTT_in_force	0.738*** (0.227)	0.837*** (0.233)	0.824*** (0.231)	0.808*** (0.228)
legal_distance_bilateral	-0.791*** (0.147)	-0.792*** (0.147)	-0.793*** (0.147)	-0.705*** (0.175)
religious_distance	-1.216*** (0.254)	-1.240*** (0.256)	-1.241*** (0.256)	-1.518*** (0.383)
cultural_distance	-0.243*** (0.057)	-0.242*** (0.057)	-0.243*** (0.057)	-0.153* (0.089)
10Y bond yield differential	0.023 (0.035)	0.034 (0.036)	0.024 (0.035)	-0.020 (0.048)
fx_volatility_12m_annual	0.122 (0.147)	0.127 (0.148)	0.147 (0.146)	0.071 (0.202)
fx_ln_change_annual	0.019 (0.034)	0.020 (0.034)	0.015 (0.035)	0.018 (0.039)
cit_signed_diff	0.020*** (0.006)	0.022*** (0.006)	0.023*** (0.006)	0.026*** (0.007)
tgt_pmr_overall	2.106*** (0.814)	1.701** (0.772)	2.228** (0.954)	2.172** (1.002)
tgt_health_exp_pct_gdp		-0.028 (0.070)	0.021 (0.076)	0.073 (0.107)
tgt_hc_openness_index			0.523 (0.462)	0.522 (0.621)
Revenue ratio				-0.061 (0.045)
Constant	-13.498 (11.246)	-12.424 (11.173)	-15.920 (11.407)	-18.604 (12.765)
Observations	34464	34218	34218	15097
Pseudo R-sq	0.473	0.474	0.474	0.500
Wald chi2	775.00	771.37	783.18	625.48
Prob > chi2	0.000	0.000	0.000	0.000

Standard errors in parentheses. Three-way FE: Year + acq_iso2 + tgt_iso2; pair clustering (acq x tgt). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.5 Robustness Checks

Product Market Regulation Identification (B9) Since my regulatory regression finding rests on *tgt_pmr_overall*, I test in Block 9 whether this

coefficient is identified from within-country or cross-country variation, by varying the fixed effects structure. Under the headline three-way specification, where target country fixed effects absorb time-invariant country levels, I find *PMR* positive and significant at $\beta = 2.172$ ($p < 0.05$), identified from within-country regulatory change. When I drop target country fixed effects (column 2), *PMR* is identified from cross-target variation and the coefficient turns negative at $\beta = -0.645$ ($p < 0.05$), under year fixed effects only (column 3) it is negative and insignificant at $\beta = -0.336$. This sign reversal shows the within-country and cross-country specifications identify two distinct margins: an increase in regulatory restrictiveness within a target country attracts inbound deal flow, while the cross-country level of *PMR* is not positively associated with deal inflows. The within-country result is the relevant test of the regulation hypothesis. I additionally find the within-country result is carried by the post-2008 period at $\beta = 2.991$ ($p < 0.01$), the pre-2008 subsample is too small to identify *PMR* ($N = 1,708$).

Product Market Regulation Decomposition (B13) To identify which component of product market regulation drives the headline regulation result, in Block 13 I decompose *tgt.pmr_overall* into four sub-indices: state distortions, barriers to entry, pharmacy-sector regulation, and retail distribution regulation (OECD 2025). Column 1 reports the headline aggregate result ($\beta = 2.172$, $p < 0.05$). In column 2 I replace the aggregate index with the four sub-indices entered jointly. I find the barriers-to-entry sub-index positive and significant at $\beta = 4.080$ ($p < 0.01$), while state distortions, pharmacy regulation, and retail distribution regulation are insignificant. The state-distortions and barriers-to-entry sub-indices are correlated at 0.88 within the target side, so their coefficients are not fully independent, the barriers-to-entry coefficient nonetheless remains significant at the one-percent level. The remaining sub-index pairs are weakly correlated (all $|r| < 0.42$). Column 3 confirms the pharmacy-sector sub-index alone is insignificant ($\beta = 1.046$). The decomposition indicates the headline product market regulation effect is carried by the barriers-to-entry component, consistent with a regulatory mechanism operating through restrictions on market entry rather than through state ownership or sector-specific regulation.

Healthcare Regulatory Regime and Brexit (B8). In Block 8 I substitute a healthcare-specific four-tier regulatory regime classification for the broad *PMR* index. Columns 1 to 3 drop target country fixed effects so the regime tier is identified across countries. With *PMR* excluded (column 2), I find *FDA*-tier and *EMA*-tier targets attract significantly higher deal volume than the lowest-tier baseline, at $\beta = 1.148$ ($p < 0.05$) and $\beta = 0.993$ ($p <$

0.01). When I add *PMR* back (column 3), the *FDA*-tier coefficient stays marginally significant at $\beta = 1.075$ ($p < 0.10$) while the *EMA*-tier coefficient loses significance at $\beta = 0.447$, *PMR* itself enters negative at $\beta = -0.568$ ($p < 0.10$), consistent with the cross-country pattern I document in Block 9. The aggregate UK post-2021 Brexit interaction is insignificant (column 4, $\beta = -0.156$). In column 5 I decompose this by acquiror regulatory tier, and I find no individually significant interaction ($\chi^2(3) = 1.96$, $p = 0.581$). I find no effect of post-Brexit regulatory divergence on cross-border UK healthcare M&A, though the two-year post-Brexit window limits the number of observations. **Healthcare Market Characterization (B11)** In Block 11 I extend the headline healthcare specification with demand-side and supply-side market characteristics. Adding the population-aged-65 share (column 2) leaves *PMR* essentially unchanged at $\beta = 2.193$ ($p < 0.05$). When the universal health coverage index enters (column 3), *PMR* loses significance at $\beta = 1.129$. I find this reflects sample composition rather than absorption: *UHC* coverage restricts the sample from 15,097 to 3,291 observations, and the *PMR* point estimate stays large in the following columns ($\beta = 2.386$ and $\beta = 2.380$, both above the headline 2.172) while the standard error roughly doubles. The significance loss is driven by reduced precision in the smaller OECD-weighted subsample, not by *UHC* displacing *PMR*. The *UHC* index itself enters negative and significant at $\beta = -0.135$ ($p < 0.01$), and physicians per 1,000 inhabitants enters marginally positive at $\beta = 0.879$ ($p < 0.10$). Health expenditure, healthcare openness, and the population-65 share are insignificant throughout. **FDI Regulatory Restrictiveness (B12)** In Block 12 I test the OECD FDI Regulatory Restrictiveness Index as an alternative regulatory measure, under four fixed effects structures and a bilateral distance form. I find target-country FDI restrictiveness negative across all specifications, consistent with the prediction that foreign-acquisition barriers deter inbound deal flow, but insignificant throughout (column 1, $\beta = -4.175$). Acquiror-country FDI restrictiveness is positive and significant under cross-acquiror identification (columns 3 and 4, $\beta = 3.6$ and $\beta = 3.3$, $p < 0.01$), but is absorbed under the three-way specification, which shows the relationship is identified from cross-country level differences rather than within-country policy variation. The headline *PMR* coefficient stays positive and marginally significant under three-way fixed effects with the FDI index included ($\beta = 1.470$, $p < 0.10$), confirming my regulation result is not sensitive to direct controls for FDI restriction. **Further robustness checks** I find the headline results unaffected by four further specifications,

reported in full in Appendix D. In Block 5 I add the real and nominal stock return differentials and split the sample at the 2008 financial crisis, the H_1 – H_5 coefficients are stable in the post-2008 subsample, while the pre-2008 subsample ($N = 1,708$) is too small for reliable identification. In Block 6 I replace the count specification of domestic M&A activity with a deal-value specification, the headline coefficients on PMR , religious distance, and the treaty indicators are unchanged. In Block 7 I measure cultural distance with the six-dimension Kogut-Singh index rather than the four-dimension index, the six-dimension measure enters negative and significant ($\beta = -0.270$, $p < 0.05$) and leaves the remaining coefficients unchanged. In Block 10 I add country-level institutional controls for disclosure quality and governance, PMR and the CIT differential remain positive and significant, and the disclosure and governance variables are identified only when country fixed effects are removed. Across all four panels the H_1 – H_5 conclusions are unchanged.

6.6 Block contribution summary

Table 9 summarises the marginal contribution of each theoretical block to model fit. To isolate the contribution of each block from changes in sample composition, the decomposition holds the sample fixed at the common $N = 15,097$ for which all variables in the full specification are non-missing. The gravity baseline accounts for the largest share of explanatory power, with a pseudo- R^2 of 0.479. Adding the institutional block raises the pseudo- R^2 to 0.490 ($\Delta R^2 = +0.012$), and the cultural block raises it to 0.496 ($\Delta R^2 = +0.006$). The valuation arbitrage block adds little explanatory power, raising the pseudo- R^2 to 0.498 ($\Delta R^2 = +0.002$), consistent with the finding that three of the four H_2 channels are insignificant and only the CIT differential carries a coefficient. The healthcare block raises the pseudo- R^2 to 0.500 ($\Delta R^2 = +0.001$). On the common sample the full cascade adds 0.021 to the pseudo- R^2 over the gravity baseline, with the institutional and cultural blocks together contributing the largest share of that gain.

Table 9: Block Contribution Summary: Nested Cascade R-sq Decomposition (Common Sample, N=15,097)

Block	Variables added	N	Pseudo R ²	ΔR^2	Wald χ^2
(1) Gravity	tgt/acq GDP, ln distance, bilateral trade, ln(1+) dom deal counts, both_eu, both_oecd	15,097	0.479	—	310.65
(2) +Institutional	BIT_in_force, DTT_in_force, legal_dist_bilat	15,097	0.490 +0.012		370.78
(3) +Cultural	religious_dist, cultural_dist	15,097	0.496 +0.006		568.03
(4) +FX/Macro	Yield Diff (acq-tgt), fx_volat_12m, fx_ln_chg, CIT diff (acq-tgt)	15,097	0.498 +0.002		600.68
(5) +Healthcare	tgt_pmr_overall, tgt_health_exp, tgt_hc_openness, Log HC Rev Ratio	15,097	0.500 +0.001		625.48
<i>Total</i> (1)→(5)	20 covariates + Constant; 3-way FE: Year + Acq + Tgt	+15,097	0.500 +0.021		625.48

Notes: Sample fixed at the N=15,097 common sample where all RHS variables in column (5) are non-missing. PPML with Year + Acq + Tgt fixed effects and pair clustering. Domestic deal counts entered as ln(1+x). Wald χ^2 tests joint significance of all included slope coefficients.

7 Discussion

In the standard cross-border M&A literature, acquirors exploit financial arbitrage opportunities across markets. In healthcare, regulatory access drives cross-border deal flow rather than financial arbitrage. This chapter develops that argument hypothesis by hypothesis, positions the findings relative to Erel et al. (2012), and then discusses two patterns that cut across the analysis, the extensive versus intensive margin and the firm-level evidence, before synthesising which theories the data support.

7.1 Gravity and the Bilateral Trade (H_1)

The gravity hypothesis is supported on its bilateral predictions and silent on its mass predictions. Geographic distance enters negative and significant throughout the cascade, and bilateral trade enters positive and significant, both consistent with the gravity tradition and with the factor-mobility logic that established economic linkage channels cross-border capital flows. The gravity block is also the single largest contributor to model fit, which confirms that the gravity framework provides the structural baseline for cross-border healthcare M&A, as it does for cross-border M&A generally in Erel et al. (2012). The GDP results require a more careful reading. Acquiror-country GDP is statistically null throughout, and target-country GDP loses signifi-

cance in the full specification. Under the three-way fixed-effects structure, acquiror and target country fixed effects absorb time-invariant and slowly-moving country-level scale, so country GDP has limited residual within-country variation through which to identify. The bilateral gravity terms, distance and trade, which vary across country pairs, survive precisely because they are not absorbed in this way. The gravity mechanism therefore operates in the data through bilateral trade channel rather than GDP channel.

7.2 Valuation Arbitrage and the Financial Channel (H_2)

The valuation arbitrage hypothesis is rejected for its financial channels and supported only for tax. The bilateral bond yield differential, foreign exchange volatility, and the exchange rate change are all statistically null. Under the Efficient Market Hypothesis, cross-border deal flow should respond to bilateral differences in the cost of capital, in the healthcare sample it does not. This is a substantive divergence from the valuation emphasis in the broader cross-border M&A literature, where equity and cost-of-capital differentials are treated as central drivers. The result suggests that cross-border healthcare M&A is not, in the main, a cost-of-capital arbitrage, the financial arbitrage channel that organises much of the general literature does not carry the healthcare-specific data. The corporate income tax differential is the single exception. It enters positive and significant, indicating that deal flow rises when the acquiror country has a higher corporate tax rate than the target country. The pattern is consistent with a tax-driven motive: acquirors domiciled in high-tax jurisdictions acquire targets in lower-tax jurisdictions, where the acquired operation's profits are taxed more favourably. This stands in mild tension with the Eclectic Paradigm as stated by Markusen (1995), who argues that tax avoidance is not a primary motive for foreign direct investment and that firms typically choose locations first and minimise tax afterward. The healthcare evidence indicates that, at the margin, the tax differential is a determinant of cross-border deal direction even if it is not the dominant one.

7.3 Institutional and Cultural Distance as Information Friction (H_3)

The institutional and cultural distance hypothesis is the most clearly supported of the five. Legal, religious, and cultural distance all enter negative and significant, and the institutional and cultural blocks together contribute the largest incremental gain to model fit beyond the gravity baseline. This is consistent with the information-asymmetry strand of the Efficient Market Hypothesis: distance along legal, religious, and cultural dimensions raises the cost of acquiring and verifying information across a country pair, and that cost deters cross-border deal flow. The finding aligns with Porta et al. (1998) on legal origin and investor protection and with Stulz and Williamson (2003) on religion and culture as predictors of foreign investment patterns. Religious distance carries the largest coefficient, followed by legal distance, with cultural distance the smallest, and the ordering is stable across the cascade. I interpret religious distance not as a literal religious mechanism but as a proxy for deep, slow-moving informational and relational gaps between societies that are not captured by the legal or Hofstede-based measures. The broader implication is that, for cross-border healthcare M&A, the friction is informational rather than financial, the distance variables, not the valuation variables, are what move deal flow once gravity is accounted for.

7.4 Market Depth, Treaties, and the Integration Reversal (H_4)

The market depth and financing hypothesis receives mixed support. Consistent with the factor-mobility logic of New Trade Theory, under which institutional reductions in investment friction channel capital flows into cross-border M&A, the double tax treaty indicator and shared OECD membership both enter positive and significant, and the acquiror's domestic M&A market depth enters positive, consistent with deeper home markets giving acquirors greater deal-making capacity and financing access. Two results run against the prediction. The bilateral investment treaty indicator is null throughout: the treaty instrument most directly aimed at protecting cross-border investment does not noticeably increase healthcare deal flow, whereas the tax treaty does. And shared EU membership enters negative and significant, deeper integration is associated with fewer cross-border deals, not more. The EU reversal is the factor-mobility logic operating in the opposite direction. Cross-border

M&A is, in this framework, a channel that substitutes for impeded goods and capital flows, where integration has already removed the barriers, as within the EU single market, the friction that acquisition exists to overcome is largely absent, and a target market can be served through trade or greenfield entry without acquiring an incumbent.

7.5 Regulation: the Barriers-to-Entry Mechanism (H_5)

The regulation hypothesis is supported, and it is the central contribution of this thesis. Target-country product market regulation enters positive and significant in the headline specification, with the largest coefficient in the model. The headline result establishes that more stringent product market regulation in a target country is associated with greater inbound cross-border healthcare M&A. This is the opposite of the intuitive prediction that regulation deters foreign entry, and the Eclectic Paradigm explains why. Regulation that raises barriers to entry is a location disadvantage to greenfield entry specifically: a foreign firm cannot easily build a presence from scratch in a heavily regulated market, because licensing, approval, and compliance infrastructure are slow and costly to assemble. The multinational enterprise overcomes this location disadvantage by internalizing an incumbent — acquiring a firm that already holds the licenses, approvals, and regulatory standing. Regulation that raises the cost of organic entry therefore raises the scarcity value of an already-compliant incumbent, and the acquisition channel becomes the profitable mode of entry. This is the location and internalization logic of Markusen (1995) applied to regulation rather than to tariffs and transport costs. The Block 9 identification test shows that this effect is identified from within-country regulatory change. Under the three-way specification, product market regulation enters positive, when target country fixed effects are dropped and the coefficient is identified from cross-country variation, the sign reverses to negative. The two specifications identify two distinct margins. The cross-country comparison conflates regulation with everything else that differs between heavily- and lightly-regulated countries, market maturity, openness, size, and on that margin highly-regulated countries do not attract more deals. The within-country margin isolates the effect of a country tightening its own regulation over time, holding the country fixed, and on that margin regulatory tightening attracts inbound deal flow. The within-country margin is the correct test of the rents mechanism, because the mechanism is about a change in entry cost, not about a cross-sectional correlation between regulation

and market characteristics. The Block 13 decomposition identifies which component of regulation drives the result, and the answer is consistent with the mechanism. Decomposing the *economy-wide* PMR index into its two sub-indices and two sectoral indicators, the barriers-to-entry sub-index carries the entire effect, while state distortions, pharmacy-sector regulation, and retail distribution regulation are all insignificant. This is direct evidence on the economic mechanism: the regulation channel operates specifically through barriers to market entry, not through state ownership of healthcare assets or through sector-specific price and licensing rules. This finding is the primary way in which the thesis extends Erel et al. (2012). The standard cross-border M&A framework is built on gravity and financial-arbitrage variables and does not consider sectoral regulation. The healthcare evidence shows that regulatory barriers to entry are significant and economically large determinant of cross-border deal flow. The channel is plausibly sharper in healthcare than in other sectors: drug approval regimes, provider licensing, and reimbursement-system entanglement make healthcare incumbents especially difficult to replicate through greenfield entry, so the acquisition-over-greenfield logic is at its strongest in this industry. The modest contribution of the regulation variable to incremental model fit, noted in the block contribution summary, does not diminish this: the variable has limited within-country variation under three-way fixed effects, and a determinant can be economically important and statistically robust without explaining a large share of variance. The contribution is the identification of a significant sectoral channel that the prior literature omits.

7.6 Bilateral Friction Cost

A pattern that cuts across the deal-count and deal-size analyses concerns the distinction between the number of deals and their average size. In the deal-count specification, geographic distance enters negative, in the deal-value specification, distance enters positive. The reversal is consistent with the firm-selection mechanism of Melitz (2003), set out in the theory chapter: bilateral friction acts as a fixed cost of transacting across a country pair, so friction reduces the number of deals that occur, while raising the average size of the deals that do, because only large transactions justify the fixed cost. The distance reversal between the two specifications is therefore not a contradiction but the Melitz selection prediction observed directly: friction screens the deal distribution, deterring marginal deals and leaving the surviving deals larger

on average.

7.7 Firm-Level Evidence and the Resource-Based View

The firm-level analysis, interpreted through the Resource-Based View, indicates that acquiror resources govern deal size. The log of acquiror total assets is the dominant firm-level determinant: larger acquirors undertake larger transactions, and acquiror size absorbs much of the explanatory content that the bilateral gravity and distance variables carried at the deal level. This is consistent with the Resource-Based View reading that the acquiror's resource base, here, scale, is a transferable advantage that enables larger cross-border transactions. These results are drawn from a restricted sample of 175 transactions with complete acquiror financials and should be read as indicative rather than conclusive, the small-sample firm-level evidence complements, but does not carry the weight of, the country-pair-year analysis.

8 Limitations

Several limitations effects the interpretation of the results. First, the analysis identifies associations rather than causal effects. The three-way fixed-effects specification absorbs time-invariant country heterogeneity and common annual shocks, and the within-country identification of the product market regulation effect rules out a class of cross-country confounders, but the design does not establish causality. The coefficients should be read as conditional associations consistent with the hypothesised mechanisms, not as causal estimates. Second, the firm-level analysis rests on a restricted sample. Complete acquiror Orbis fundamentals are available for only 175 transactions, so the firm-level results are indicative rather than conclusive and carry less weight than the country-pair-year analysis. The country-pair-year panel, which supports the main findings, is not subject to this constraint. Third, the identification of the healthcare channels is limited by the within-country variation available under the fixed-effects structure. Product market regulation is identified because it changes within countries through regulatory reform waves, but the remaining healthcare variables, health expenditure, healthcare openness, and the revenue ratio, are slow-moving and have limited within-country variation once target country fixed effects are included. Their statistical insignificance therefore reflects the limits of within-country identification as much as the absence of

an economic relationship, and should not be read as strong evidence against those channels. Fourth, the decomposition of product market regulation into its sub-indices is not perfectly clean. The state-distortions and barriers-to-entry sub-indices are correlated at 0.88, so although the barriers-to-entry component carries the effect at the one-percent level, its coefficient is not fully independent of the state-distortions component. The decomposition is therefore strong but indicative evidence on the operative regulatory dimension. Fifth, the test of post-Brexit regulatory divergence is underpowered. The post-2021 observation window spans only two years, which limits the statistical power of the difference-in-differences specification, the null result for the Brexit interaction should be read in light of this short window rather than as definitive evidence of no effect. Sixth, the analysis involves data-construction choices. Sovereign bond yields are imputed for a small number of countries, exchange-rate channels are recoded for Argentina due to hyperinflation, and the product market regulation sub-indices are drawn from OECD. The deal-level sample is restricted to transactions with disclosed deal value, which biases that sample toward larger deals. Finally, the findings are specific to one sector and one methodological framework. The thesis examines cross-border healthcare M&A through a gravity framework anchored on Erel et al. (2012), and the central result, that regulatory barriers to entry drive cross-border deal flow, is established for healthcare. Whether the regulation channel generalizes to other heavily regulated sectors is not tested here and is left to future research.

9 Conclusion

This thesis has examined what determines cross-border mergers and acquisitions within the healthcare industry. Applying a gravity framework anchored on Erel et al. (2012) to a directed bilateral panel of cross-border healthcare transactions, and estimating it by Poisson pseudo-maximum-likelihood with three-way fixed effects, I tested five hypotheses drawn from neoclassical and new trade theory, the Efficient Market Hypothesis, and the Eclectic Paradigm. The results give a clear answer. The gravity framework provides the structural baseline: bilateral distance deters cross-border healthcare deal flow and bilateral trade promotes it, and the gravity block accounts for the largest share of explanatory power. Legal, religious, and cultural distance each reduce deal flow, and together form the largest incremental determinant beyond

gravity. The financial arbitrage channels, cost-of-capital and exchange-rate differences, are not supported in this healthcare sample, only the corporate income tax differential carries a effect. Institutional integration raises deal flow through tax treaties and shared OECD membership, but not within the EU, where it reverses. The central finding concerns regulation. Target-country product market regulation is positively associated with inbound cross-border healthcare M&A, the decomposition of the regulation index shows that this effect runs specifically through the barriers-to-entry component rather than through state ownership or sector-specific regulation. The contribution of this thesis is to show that industry matters for the cross-border M&A literature. Where the literature explains deal flow through gravitational and financial channels, the evidence shows that in a single heavily-regulated sector, it is regulatory access, not financial arbitrage, that drives cross-border deal flow.

10 Future Research

The findings and limitations of this thesis suggest several directions for future research. First, the central result, that regulatory barriers to entry drive cross-border deal flow, is established for healthcare, but the underlying mechanism is not unique to it. Energy, finance, and telecommunications are likewise heavily regulated sectors in which an incumbent's regulatory standing is costly to replicate through greenfield entry. Applying the framework to other regulated sectors would establish whether the regulation channel is a healthcare-specific feature or a general characteristic of cross-border M&A in regulated industries. Second, the decomposition of product market regulation could be pursued with detailed sub-indicator coverage. A study with broader country and time coverage of the OECD product market regulation sub-indices, or with healthcare-specific regulatory sub-measures, could isolate the regulatory dimension more cleanly.

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Appendix A

This section illustrates demographic data, top 20 causes of death globally in 2000, top 20 causes of death globally in 2021, healthcare expenditure as a share of GDP by region.

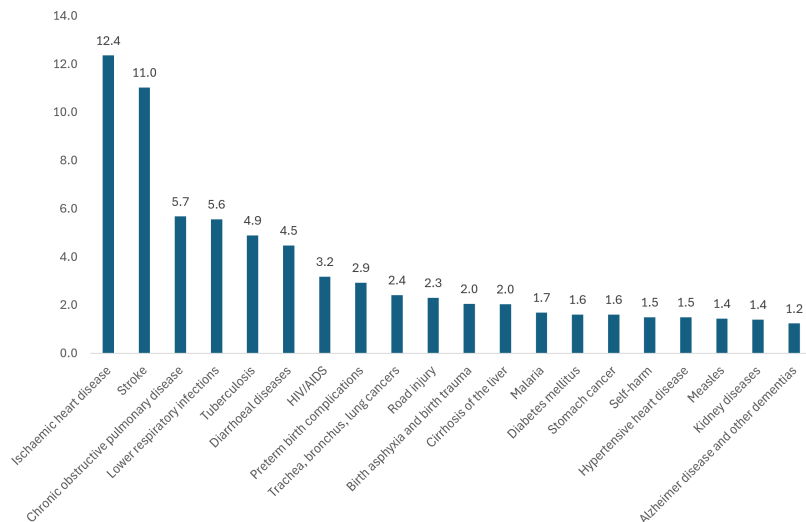


Figure 4: Global distribution of the top 20 causes of death, 2000.

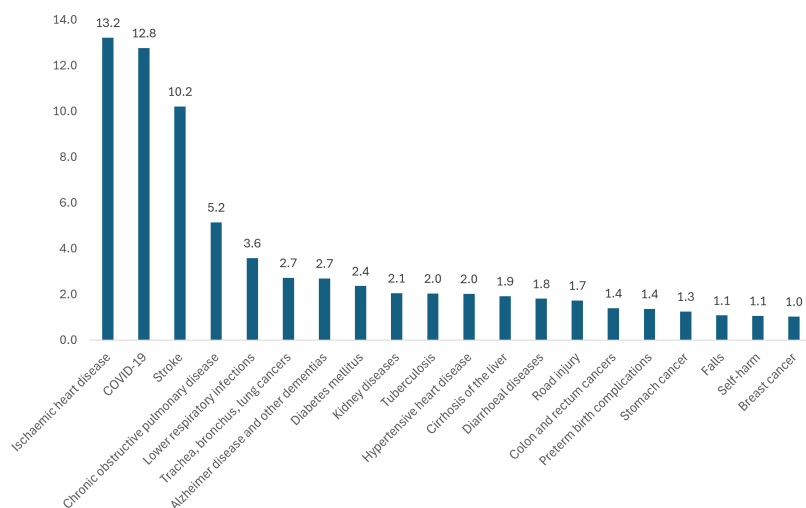


Figure 5: Global distribution of the top 20 causes of death, 2021.

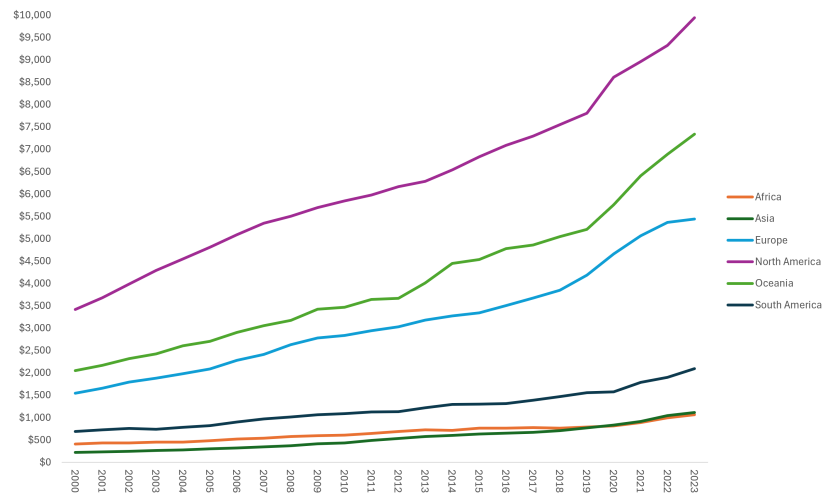


Figure 6: Healthcare expenditure as a share of GDP by region.

Appendix B

[H]

Table 10: Mapping of variables to theoretical channels

Role	Variable	Construction	Sign	Theoretical Interpretation	Data Source	Citation
Panel A. Neoclassical & New Trade Theory (gravity, frictions, selection)						
Mass (gravity)	acq_ln_total_gdp	ln(acquiror total GDP, USD)	+	Acquiror-country economic mass; pool of potential acquirors	World Bank WDI	Tinbergen (1962); Anderson & van Wincoop (2003)
Mass (gravity)	tgt_ln_total_gdp	ln(target total GDP, USD)	+	Target-country economic mass; pool of potential targets	World Bank WDI	Tinbergen (1962); Anderson & van Wincoop (2003)
Friction (dist)	ln_distance_km	ln(great-circle km between capitals)/+		Geographic friction; Melitz selection on extensive vs. intensive margin	mapsofworld	Erel, Liao & Weisbach (2012); Melitz (2003)
Friction (trade)	ln_bilateral_trade	ln(bilateral exports + imports, USD)		Pre-existing trade linkages reduce information frictions	UN Comtrade	Head & Mayer (2014)
Integration	both_eu	$1\{\text{acq EU} = 1 \wedge \text{tgt EU} = 1\}$ at deal year	+	EU single market lowers transaction costs	EU accession dates	Erel, Liao & Weisbach (2012)
Integration	both_oecd	$1\{\text{acq OECD} = 1 \wedge \text{tgt OECD} = 1\}$ at deal year	+	OECD shared regulatory frameworks reduce frictions	OECD accession dates	Erel, Liao & Weisbach (2012)
Treaty (Inv)	BIT_in_force	$1\{\text{BIT entry-into-force deal year}\}$	$\leq$	Bilateral investment treaty lowers expropriation risk	UNCTAD IIA	di Giovanni (2005)
Treaty (Tax)	DTT_in_force	$1\{\text{DTT entry-into-force deal year}\}$	$\leq$	Double-tax treaty eliminates double taxation	IBFD/OECD	di Giovanni (2005)
Panel B. Eclectic Paradigm (OLI) – Location pillar						
L-Reg. Rents	tgt_pmr_overall	OECD PMR overall score, target country, deal year	+	Regulated markets protect post-acquisition margins	OECD PMR	Dunning (1977, 1988)
L-Market Size	health_exp_pct_gdp	Current health expenditure (% of GDP), target country	+	Healthcare demand depth in target country	World Bank WDI	Dunning (1988)

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Table 10 continued from previous page

Role	Variable	Construction	Sign	Theoretical Interpretation	Data Source	Citation
L-Openness	hc_openness_index	OECD FDI RRI, health subsector, target country	+	Healthcare-specific FDI policy openness	OECD FDI RRI	Kalinova et al. (2010)
L-Rel. Scale	revenue_ratio	$\ln(\text{acq revenue}/\text{tgt revenue})$	+	Relative scale of HC sector across acquiror and target	Compustat	Dunning (1988)
L-Reg. Regime	tgt_reg_tier	Four-tier target-country pharmaceutical regulatory regime (1=FDA ... 4=Other)	+	Stricter regime raises the rent to an already-compliant incumbent	Author's classification	Erel, Liao & Weisbach (2012)
Panel C. Eclectic Paradigm (OLI O/I) and Resource-Based View						
O-Endowment	Acq_LogTA	$\ln(\text{acquiror total assets, USD})$ at $t - 1$	+	Larger firms have more advantages to exploit abroad	O- Orbis	Dunning (1988); Barney (1991)
O-Capability	ebitda_margin_pre	EBITDA / revenue at $t - 1$, winsorised 1/99	+	Firm-specific productivity; managerial capability proxy	Orbis	Melitz (2003); Helpman et al. (2004)
RBV-Slack	leverage_pre	Total debt / total assets at $t - 1$	−	Financial-resource constraint on deal capacity	Orbis	Barney (1991)
Experience	prior_total_deals	Count of acquiror's historical M&A deals up to $t - 1$	+	Accumulated deal-execution capability	Orbis	Buckley & Casson (1976); Teece et al. (1997)
Experience	serial_acquirer	$1\{\text{acquiror prior deals} \geq 3\}$	+	Accumulated acquisition capability; serial acquiror	Orbis	Teece et al. (1997)
RBV-Maturity	acq_age_at_deal	Deal year − acquiror founding year	+	Organizational maturity / accumulated routines	Orbis	Penrose (1959); Barney (1991)
Panel D. Information Asymmetry & Distance Frictions						
Cultural Dist	cultural_distance	KS index: $\frac{1}{4} \sum_k (I_k^a - I_k^t)^2 / V_k$	−	Kogut-Singh composite; integration friction	Hofstede	Kogut & Singh (1988); Hofstede (1980)
Religious Dist	religious_distance	Squared Euclidean distance over religion-share vectors	−	Religious-composition dissimilarity (trust friction)	Our World in Data	Stulz & Williamson (2003)

Continued on next page...

Table 10 continued from previous page

Role	Variable	Construction	Sign	Theoretical Interpretation	Data Source	Citation
Legal Distance	<code>legal_distance</code>	Distance over La Porta legal-origin classification	−	Legal-origin dissimilarity; investor-protection friction	JuriGlobe (Univ. Ottawa)	La Porta et al. (1998)
Linguistic Prox	<code>common_language</code>	$1\{\text{shared official language}\}$	+	Reduces communication / due-diligence cost	Dow & Karunaratna (2006)	Dow & Karunaratna (2006)
Industry Prox	<code>same_naics4</code>	$1\{\text{acq NAICS-4} = \text{tgt NAICS-4}\}$	+	Same NAICS-4 industry; synergy / information channel	Orbis M&A	Akerlof (1970); Ahern et al. (2015)
Panel E. Efficient Market Hypothesis – Valuation & Cost-of-Capital						
Cost of Cap.	<code>bond_yield_diff</code>	$\text{acq_yield} - \text{tgt_yield}$ (10y govt bond)	−	Cheaper home financing drives outbound M&A	OECD	Froot & Stein (1991)
Tax Arb.	<code>cit_signed_diff</code>	Acquiror – target statutory corporate income tax rate	+	High-tax acquirors acquire lower-tax targets to reduce effective tax burden	OECD Corporate Tax Rates	Erel, Liao & Weisbach (2012)
Equity Val.	<code>stock_return_diff</code>	Acquiror – target annual real equity return	+	Overvaluation hypothesis funds foreign acquisitions	Compustat	Erel, Liao & Weisbach (2012)
Currency Vol.	<code>fx_volatility_12m</code>	Annualised SD of monthly bilateral ln FX return	−	FX uncertainty deters cross-border deal-making	Compustat	Froot & Stein (1991)
Currency Chg.	<code>fx_ln_change_annual</code>	$\ln(S_{ij,t}/S_{ij,t-1})$, bilateral nominal FX rate	+	Target-currency depreciation lowers the effective price of target assets	Compustat	Froot & Stein (1991)

Appendix C

This section displays the cross-border deal volume figures and the deal matrix from Erel et al. (2012), which have been reconstructed in Figure 2 and Table 1 of this paper, respectively.

Figure 7: Original Cross-Border Deal Volume (Erel et al., 2012)

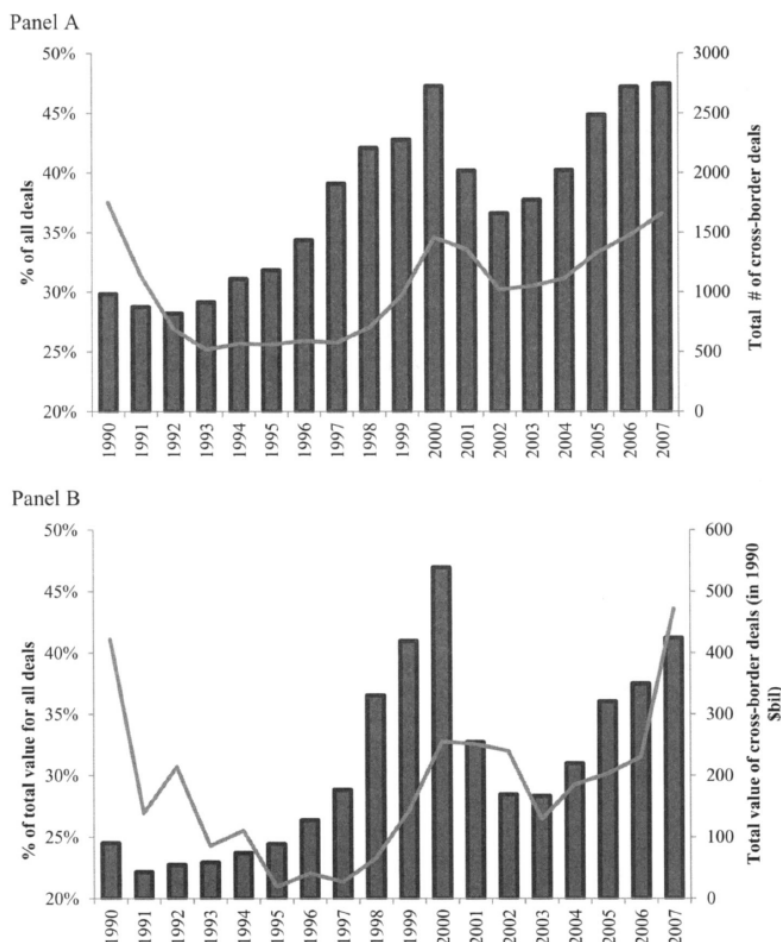


Figure 1. Total value of cross-border mergers and acquisitions. This figure plots the number (ratio) (Panel A) and the value (ratio) (Panel B) of cross-border deals with deal value larger than \$1 million between 1990 and 2007. Bars represent numbers or values in a given year while solid lines represent the fraction of cross-border acquisitions relative to the total number or deal value of all acquisitions in a given year, including domestic ones. All values are in 1990 dollars.

Source: Erel et al. (2012)

Appendix D

Table 11: B5: Robustness Panel (PPML, Year + Acq + Tgt FE)

	(1) Anchor	(2) +Real Stock	(3) +Nom Stock	(4) Pre-2008	(5) Post-2008
tgt_ln_total_gdp	0.404 (0.308)	0.367 (0.324)	0.361 (0.326)	-0.649 (1.154)	0.365 (0.463)
acq_ln_total_gdp	-0.421 (0.362)	-0.452 (0.375)	-0.460 (0.375)	3.708*** (1.239)	-1.450*** (0.561)
ln_distance_km	-0.298** (0.138)	-0.260* (0.141)	-0.260* (0.141)	0.442 (0.418)	-0.483*** (0.143)
ln_bilateral_trade	0.604*** (0.119)	0.650*** (0.122)	0.649*** (0.122)	1.117*** (0.377)	0.472*** (0.129)
Log(1 + tgt domestic deal count)	0.046 (0.079)	0.046 (0.078)	0.045 (0.079)	0.196 (0.181)	-0.003 (0.085)
Log(1 + acq domestic deal count)	0.235*** (0.085)	0.216** (0.086)	0.219** (0.086)	0.265 (0.236)	0.096 (0.100)
both_eu	-0.419* (0.250)	-0.391 (0.253)	-0.389 (0.253)	0.171 (0.619)	-0.334 (0.266)
both_oecd	1.201*** (0.292)	1.384*** (0.391)	1.380*** (0.390)	0.892 (1.467)	0.945*** (0.291)
BIT_in_force	-0.234 (0.198)	-0.265 (0.208)	-0.267 (0.208)	-0.556 (0.787)	-0.202 (0.195)
DTT_in_force	0.808*** (0.228)	0.694*** (0.243)	0.693*** (0.242)	-0.680 (1.150)	0.945*** (0.234)
legal_distance_bilateral	-0.705*** (0.175)	-0.693*** (0.174)	-0.692*** (0.175)	-0.422 (0.395)	-0.706*** (0.186)
religious_distance	-1.518*** (0.383)	-1.468*** (0.423)	-1.470*** (0.423)	0.235 (1.584)	-1.555*** (0.341)
cultural_distance	-0.153* (0.089)	-0.133 (0.096)	-0.133 (0.096)	-0.256 (0.343)	-0.185** (0.094)
10Y bond yield differential (acq - tgt)	-0.020 (0.048)	-0.035 (0.049)	-0.033 (0.049)	0.347 (0.224)	-0.040 (0.050)
fx_volatility_12m_annual	0.071 (0.202)	0.111 (0.203)	0.105 (0.202)	0.163 (0.334)	-0.187 (0.325)
fx_ln_change_annual	0.018 (0.039)	0.017 (0.039)	0.017 (0.039)	0.038 (0.076)	0.064 (0.071)
cit_signed_diff	0.026*** (0.007)	0.024*** (0.007)	0.024*** (0.007)	-0.080* (0.047)	0.026*** (0.009)
tgt_health_exp_pct_gdp	0.073 (0.107)	0.087 (0.109)	0.089 (0.109)	-1.152*** (0.354)	0.217* (0.123)
tgt_pmr_overall	2.172** (1.002)	2.308** (1.025)	2.331** (1.038)	0.000 (.)	2.991*** (1.025)
tgt_hc_openness_index	0.522 (0.621)	0.549 (0.651)	0.551 (0.651)	3.349 (2.631)	0.524 (0.868)
Log healthcare service revenue ratio (acq/tgt)	-0.061 (0.045)	-0.066 (0.049)	-0.066 (0.049)	-0.009 (0.075)	-0.026 (0.058)
stock_return_diff_real		-0.046 (0.066)			
Nominal stock market return diff (acq - tgt)			-0.034 (0.108)		
Constant	-18.604 (12.765)	-18.607 (13.031)	-18.227 (13.056)	-110.509** (53.547)	14.474 (18.599)
Observations	15097	13790	13790	1708	11354
Pseudo R-sq	0.500	0.500	0.500	0.504	0.494
Wald chi2	625.48	596.31	597.06	180.23	588.91
Prob > chi2	0.000	0.000	0.000	0.000	0.000

Standard errors in parentheses. Three-way FE: Year + acq_iso2 + tgt_iso2; pair clustering (acq x tgt). col1: headline B1 col5 reference. col2: adds stock_return_diff_real (real return differential). Col3: adds nominal acq+tgt stock returns separately. col4-5: temporal sample splits at GFC. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 12: Block 6: Domestic Activity & Stock Return Robustness (PPML, Year + Acq + Tgt FE)

	(1) Count + Real	(2) Value + Real	(3) Count + Nom	(4) Value + Nom
tgt_ln_total_gdp	0.367 (0.324)	0.150 (0.371)	0.361 (0.326)	0.147 (0.378)
acq_ln_total_gdp	-0.452 (0.375)	-0.797** (0.399)	-0.460 (0.375)	-0.795** (0.400)
ln_bilateral_trade	0.650*** (0.122)	0.665*** (0.121)	0.649*** (0.122)	0.665*** (0.121)
ln_distance_km	-0.260* (0.141)	-0.237* (0.139)	-0.260* (0.141)	-0.237* (0.139)
Log(1 + tgt domestic deal count)	0.046 (0.078)		0.045 (0.079)	
Log(1 + acq domestic deal count)	0.216** (0.086)		0.219** (0.086)	
both_eu	-0.391 (0.253)	-0.332 (0.255)	-0.389 (0.253)	-0.332 (0.255)
both_oecd	1.384*** (0.391)	1.408*** (0.422)	1.380*** (0.390)	1.407*** (0.421)
BIT_in_force	-0.265 (0.208)	-0.238 (0.215)	-0.267 (0.208)	-0.238 (0.215)
DTT_in_force	0.694*** (0.243)	0.739*** (0.257)	0.693*** (0.242)	0.739*** (0.256)
legal_distance_bilateral	-0.693*** (0.174)	-0.672*** (0.175)	-0.692*** (0.175)	-0.672*** (0.175)
religious_distance	-1.468*** (0.423)	-1.351*** (0.524)	-1.470*** (0.423)	-1.351*** (0.525)
cultural_distance	-0.133 (0.096)	-0.139 (0.104)	-0.133 (0.096)	-0.139 (0.104)
10Y bond yield differential (acq - tgt)	-0.035 (0.049)	-0.044 (0.051)	-0.033 (0.049)	-0.044 (0.050)
fx_volatility_12m_annual	0.111 (0.203)	0.104 (0.212)	0.105 (0.202)	0.104 (0.211)
fx_ln_change_annual	0.017 (0.039)	0.039 (0.043)	0.017 (0.039)	0.039 (0.043)
cit_signed_diff	0.024*** (0.007)	0.024*** (0.007)	0.024*** (0.007)	0.024*** (0.007)
stock_return_diff_real	-0.046 (0.066)	-0.001 (0.070)		
tgt_health_exp_pct_gdp	0.087 (0.109)	0.182 (0.125)	0.089 (0.109)	0.182 (0.125)
tgt_pmr_overall	2.308** (1.025)	2.355** (1.017)	2.331** (1.038)	2.355** (1.023)
tgt_hc_openness_index	0.549 (0.651)	0.456 (0.637)	0.551 (0.651)	0.453 (0.642)
Log healthcare service revenue ratio (acq/tgt)	-0.066 (0.049)	-0.070 (0.051)	-0.066 (0.049)	-0.070 (0.052)
tgt_ln_domestic_deal_value		0.048 (0.058)		0.048 (0.058)
acq_ln_domestic_deal_value		0.145*** (0.044)		0.145*** (0.045)
Nominal stock market return diff (acq - tgt)			-0.034 (0.108)	0.015 (0.152)
Constant	-18.607 (13.031)	-6.476 (13.847)	-18.227 (13.056)	-6.435 (13.909)
Observations	13790	9118	13790	9118
Pseudo R-sq	0.500	0.478	0.500	0.478
Wald chi2	596.31	510.63	597.06	510.96
Prob > chi2	0.000	0.000	0.000	0.000

Standard errors in parentheses. Three-way FE: Year + acq_iso2 + tgt_iso2; pair clustering (acq x tgt). col1: domestic activity = count, equity channel = real return diff (preferred). col2: domestic activity = value (ELW-style), equity = real return diff. Col3: count + nominal stock return differential. Col4: value + nominal stock return differential (ELW-style). Stable coefficients on religious_distance, BIT, OECD, PMR across cols = robust to specification choice. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 13: B8: Healthcare Regulatory Regime Tier Robustness (PPML)

	(1) PMR	(2) Regm	(3) PMR + Regime	(4) Brexit DiD	(5) Brexit
tgt_ln_total_gdp	0.063 (0.154)	-0.060 (0.141)	0.021 (0.143)	0.396 (0.307)	0.394 (0.306)
acq_ln_total_gdp	-0.328 (0.346)	-0.366 (0.330)	-0.259 (0.344)	-0.414 (0.363)	-0.408 (0.363)
ln_distance_km	-0.233* (0.126)	-0.195 (0.132)	-0.250* (0.128)	-0.302** (0.137)	-0.292** (0.139)
ln_bilateral_trade	0.468*** (0.122)	0.510*** (0.119)	0.470*** (0.120)	0.603*** (0.119)	0.604*** (0.119)
tgt_domestic_deal_count	0.206*** (0.061)	0.202*** (0.067)	0.149** (0.063)	0.045 (0.078)	0.045 (0.078)
acq_domestic_deal_count	0.258*** (0.082)	0.277*** (0.082)	0.260*** (0.083)	0.234*** (0.085)	0.234*** (0.086)
both_eu	-0.085 (0.227)	-0.386 (0.261)	-0.381 (0.260)	-0.435* (0.255)	-0.408 (0.269)
both_oecd	0.273 (0.344)	0.581** (0.276)	0.272 (0.331)	1.191*** (0.293)	1.189*** (0.293)
BIT_in_force	-0.286 (0.231)	-0.155 (0.214)	-0.363* (0.216)	-0.238 (0.198)	-0.236 (0.197)
DTT_in_force	0.452* (0.257)	0.459** (0.231)	0.440* (0.257)	0.806*** (0.228)	0.806*** (0.228)
legal_distance	-0.706*** (0.193)	-0.715*** (0.203)	-0.717*** (0.198)	-0.706*** (0.175)	-0.710*** (0.177)
religious_distance	-1.354*** (0.348)	-1.542*** (0.344)	-1.663*** (0.360)	-1.516*** (0.384)	-1.517*** (0.383)
cultural_distance	-0.322*** (0.082)	-0.264*** (0.079)	-0.286*** (0.081)	-0.155* (0.089)	-0.156* (0.089)
10Y_bond_yield_differential	-0.114*** (0.033)	-0.106*** (0.035)	-0.096*** (0.035)	-0.021 (0.049)	-0.021 (0.048)
fx_volatility_12m_annual	0.055 (0.219)	0.092 (0.207)	0.035 (0.217)	0.071 (0.202)	0.073 (0.202)
fx_ln_change_annual	0.014 (0.036)	0.019 (0.034)	0.017 (0.036)	0.018 (0.039)	0.018 (0.039)
cit_signed_diff	0.023*** (0.007)	0.022*** (0.007)	0.021*** (0.007)	0.026*** (0.007)	0.026*** (0.007)
tgt_health_exp_pct_gdp	-0.047 (0.037)	-0.080* (0.042)	-0.118** (0.049)	0.088 (0.114)	0.091 (0.113)
tgt_hc_openness_index	0.504* (0.291)	0.460 (0.295)	0.373 (0.314)	0.551 (0.636)	0.557 (0.636)
Revenue_ratio	-0.047 (0.029)	-0.045 (0.029)	-0.051* (0.030)	-0.061 (0.045)	-0.061 (0.045)
tgt_pmr_overall	-0.645** (0.276)		-0.568* (0.298)	2.195** (1.007)	2.186** (1.006)
FDA		1.148** (0.553)	1.075* (0.610)		
EMA		0.993*** (0.341)	0.447 (0.437)		
OtherMajor		0.426 (0.358)	0.043 (0.435)		
UK x post-2021 (Brexit reg divergence DiD)				-0.156 (0.263)	
UK x post-2021 x FDA-tier acquirer (US)					-0.082 (0.328)
UK x post-2021 x EMA-tier acquirer					0.011 (0.474)
UK x post-2021 x Other Major-tier acquirer					-0.568 (0.428)
Constant	-2.978 (9.538)	-1.385 (9.323)	-3.117 (9.609)	-18.698 (12.793)	-18.919 (12.846)
Observations	16435	17059	16435	15097	15097
Pseudo R-sq	0.487	0.480	0.489	0.500	0.500
Wald chi2 (overall)	692.09	839.08	776.42	639.39	785.14
Prob > chi2 (overall)	0.000	0.000	0.000	0.000	0.000
Brexit joint chi2(3)					1.96
Brexit joint p-value					0.581

Standard errors in parentheses. Cols (1)-(3): Year + acq_iso2 FE only (tgt FE dropped so regime tier identifies across countries). Col (4): full Year + Acq + Tgt FE (3-way); regime tier absorbed by tgt FE; UK x post-2021 captures aggregate Brexit DiD. Col (5): full 3-way FE + Brexit DiD interacted with acquirer-side regime tier (baseline = Other-tier acquirer). Brexit joint test (col 5 only): Wald chi-squared test that all three Brexit-x-acquirer-tier interactions are zero. Stars on chi2 reflect joint test p-value. Regime tier baseline: Tier 4 (Other). FDA=US; EMA=EU/EEA + UK_i=2020; Other Major=CH, JP, CA, AU, NZ, KR, SG, IL, HK + UK_i=2021. Pair clustering (acq x tgt) throughout. ln_hc_svc_revenue_ratio sample restricts N to 13-15k. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14: Correlation Matrix: Target-Country PMR Sub-Indices

	(1)	(2)	(3)	(4)	(5)
(1) PMR overall	1.000				
(2) State distortions	0.973	1.000			
(3) Barriers to entry	0.966	0.881	1.000		
(4) Pharmacy regulation	0.127	0.107	0.142	1.000	
(5) Retail distribution	0.376	0.372	0.357	0.396	1.000

Notes: Pairwise correlations of target-country product market regulation variables, country-pair-year sample ($N = 42,864$, offshore-excluded). Variable (1) is the aggregate OECD-WBG product market regulation index; variables (2)–(5) are its sub-components. The high correlation between the aggregate index and the state-distortions and barriers-to-entry sub-indices ($r \approx 0.97$) is why the aggregate is not entered jointly with its components; the decomposition in Table 13 enters the sub-indices in place of the aggregate.

Table 15: B9: PMR Identification Strategy Robustness (PPML)

	(1) 3-way FE No	(2) Tgt FE Year	(3) FE only	(4) Pre-2008	(5) Post-2008
tgt_ln_total_gdp	0.404 (0.308)	0.063 (0.154)	0.159 (0.241)	-0.649 (1.154)	0.365 (0.463)
acq_ln_total_gdp	-0.421 (0.362)	-0.328 (0.346)	-0.266 (0.198)	3.708*** (1.239)	-1.450*** (0.561)
ln_distance_km	-0.298** (0.138)	-0.233* (0.126)	-0.246 (0.201)	0.442 (0.418)	-0.483*** (0.143)
ln_bilateral_trade	0.604*** (0.119)	0.468*** (0.122)	0.446* (0.228)	1.117*** (0.377)	0.472*** (0.129)
Log(1 + tgt domestic deal count)	0.046 (0.079)	0.206*** (0.061)	0.316*** (0.065)	0.196 (0.181)	-0.003 (0.085)
Log(1 + acq domestic deal count)	0.235*** (0.085)	0.258*** (0.082)	0.397*** (0.069)	0.265 (0.236)	0.096 (0.100)
both_eu	-0.419* (0.250)	-0.085 (0.227)	-0.495** (0.246)	0.171 (0.619)	-0.334 (0.266)
both_oecd	1.201*** (0.292)	0.273 (0.344)	0.605** (0.271)	0.892 (1.467)	0.945*** (0.291)
BIT_in_force	-0.234 (0.198)	-0.286 (0.231)	-0.559** (0.232)	-0.556 (0.787)	-0.202 (0.195)
DTT_in_force	0.808*** (0.228)	0.452* (0.257)	0.232 (0.223)	-0.680 (1.150)	0.945*** (0.234)
legal_distance_bilateral	-0.705*** (0.175)	-0.706*** (0.193)	-0.776*** (0.213)	-0.422 (0.395)	-0.706*** (0.186)
religious_distance	-1.518*** (0.383)	-1.354*** (0.348)	-2.200*** (0.313)	0.235 (1.584)	-1.555*** (0.341)
cultural_distance	-0.153* (0.089)	-0.322*** (0.082)	-0.170 (0.106)	-0.256 (0.343)	-0.185** (0.094)
10Y bond yield differential (acq - tgt)	-0.020 (0.048)	-0.114*** (0.033)	-0.125*** (0.033)	0.347 (0.224)	-0.040 (0.050)
fx_volatility_12m_annual	0.071 (0.202)	0.055 (0.219)	-0.082 (0.244)	0.163 (0.334)	-0.187 (0.325)
fx_ln_change_annual	0.018 (0.039)	0.014 (0.036)	-0.008 (0.041)	0.038 (0.076)	0.064 (0.071)
cit_signed_diff	0.026*** (0.007)	0.023*** (0.007)	0.017** (0.008)	-0.080* (0.047)	0.026*** (0.009)
tgt_health_exp_pct_gdp	0.073 (0.107)	-0.047 (0.037)	-0.033 (0.042)	-1.152*** (0.354)	0.217* (0.123)
tgt_pmr_overall	2.172** (1.002)	-0.645** (0.276)	-0.336 (0.253)	0.000 (.)	2.991*** (1.025)
tgt_hc_openness_index	0.522 (0.621)	0.504* (0.291)	0.432 (0.290)	3.349 (2.631)	0.524 (0.868)
Log healthcare service revenue ratio (acq/tgt)	-0.061 (0.045)	-0.047 (0.029)	0.032 (0.027)	-0.009 (0.075)	-0.026 (0.058)
Constant	-18.604 (12.765)	-2.978 (9.538)	-8.655* (5.188)	-110.509** (53.547)	14.474 (18.599)
Observations	15097	16435	18476	1708	11354
Pseudo R-sq	0.500	0.487	0.460	0.504	0.494
Wald chi2	625.48	692.09	698.28	180.23	588.91
Prob χ^2	0.000	0.000	0.000	0.000	0.000

Standard errors in parentheses. Col (1): Year + Acq + Tgt FE (headline; PMR identified from within-country time variation). (2): Year + Acq FE only (target-country FE dropped; PMR identified from cross-target variation). Col (3): Year FE only (no country FE; PMR identified from pure cross-sectional + time variation). Col (4)-(5): Temporal sample split at 2008 GFC; 3-way FE retained. Pair clustering (acq x tgt) throughout. tgt_pmr_overall = OECD PMR overall index, target country, deal year. Within-country positive coefficient (col 1) is the relevant test of the rents-from-regulation hypothesis. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 16: B10: Country-Level Institutional Variables Identification Panel (PPML)

	(1) 3-way FE	(2) No Tgt FE	(3) Year FE only	(4) No Acq FE	(5) Distances
acq_disclosure_index	3.636 (3.118)	4.060 (3.235)	2.064*** (0.451)	2.031*** (0.423)	
tgt_disclosure_index	2.916** (1.364)	-0.387 (0.463)	-0.637 (0.497)	3.292** (1.324)	
acq_wgi_composite	0.413 (0.292)	0.347 (0.224)	0.912*** (0.315)	0.854** (0.365)	
tgt_wgi_composite	0.267 (0.400)	-0.253 (0.250)	-0.102 (0.295)	0.151 (0.439)	
cit_signed_diff	0.025*** (0.007)	0.024*** (0.007)	0.019*** (0.007)	0.013* (0.007)	0.025*** (0.007)
Bilateral disclosure distance					-0.926 (0.597)
Bilateral WGI composite distance					-0.235 (0.255)
tgt_pmr_overall	1.572 (1.020)	-0.688** (0.283)	-0.347 (0.252)	1.523 (1.124)	2.094** (0.993)
Observations	15097	16435	18476	16992	15097
Pseudo R-sq	0.501	0.488	0.472	0.486	0.500
Wald chi2	621.68	750.62	1318.06	926.62	614.85
Prob > chi2	0.000	0.000	0.000	0.000	0.000

Standard errors in parentheses. Col (1): 3-way FE (Year + Acq + Tgt). Country-level disclosure and WGI levels absorbed by FE. Col (2): Year + Acq FE only; target-country levels identify off cross-target variation. Col (3): Year FE only; both acq and tgt levels identify (ELW-style identification). Col (4): Year + Tgt FE only; acquirer-country levels identify off cross-acquirer variation. Col (5): 3-way FE with disclosure and WGI in bilateral distance form —acq - tgt—. `cit_signed_diff` = `acq_cit_rate` - `tgt_cit_rate` is in the headline spec (FULSPEX) and identifies under all FE structures. Table shows institutional variables + `tgt_pmr_overall` (stability anchor); gravity controls suppressed. Pair clustering (acq x tgt) throughout. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 17: B11: Healthcare Market Characterization Robustness (PPML, Year + Acq + Tgt FE)

	(1) Headline	(2) +Pop65	(3) +UHC	(4) +Physicians	(5) Alt Spec
tgt_pmr_overall	2.172** (1.002)	2.193** (1.012)	1.129 (1.638)	2.386 (1.749)	2.380 (1.731)
tgt_health_exp_pct_gdp	0.073 (0.107)	0.085 (0.103)	0.079 (0.172)	0.039 (0.210)	
tgt_hc_openness_index	0.522 (0.621)	0.553 (0.623)	0.592 (0.644)	0.965 (1.000)	0.871 (0.788)
ln_hc_svc_revenue_ratio	-0.061 (0.045)	-0.059 (0.046)	0.079 (0.109)	0.020 (0.122)	0.020 (0.122)
tgt_pop_65_pct		0.048 (0.114)	-0.091 (0.199)	-0.078 (0.196)	-0.091 (0.191)
tgt_uhc_index			-0.102** (0.050)	-0.135*** (0.052)	-0.135*** (0.051)
tgt_physicians_per_1k				0.864* (0.521)	0.879* (0.498)
Observations	15097	15097	3291	3030	3030
Pseudo R-sq	0.500	0.500	0.514	0.482	0.482
Wald chi2	625.48	631.59	335.54	336.91	336.00
Prob χ^2	0.000	0.000	0.000	0.000	0.000

Standard errors in parentheses Col (1): Headline B1 col 5 reference (target-side healthcare characteristics). Col (2): + Target country share of population aged 65+ (demographic demand driver). Col (3): + Universal Health Coverage index (public-system quality measure). Col (4): + Physicians per 1,000 inhabitants (supply-side healthcare infrastructure density). Col (5): Alternative characterization — replaces health_exp_pct_gdp with the three demand/supply variables. Three-way FE: Year + Acq + Tgt; pair clustering throughout. Sample: cols 1-2 use full N=15,097; cols 3-5 restricted to UHC+physicians coverage (N 3,000, OECD-weighted). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 18: B12: FDI Regulatory Restrictiveness Index Identification Panel (PPML)

	(1) 3-way FE	(2) No Tgt FE	(3) Year FE only	(4) No Acq FE	(5) Distance
acq_fdi_total	4.981 (4.173)	4.584 (4.331)	3.614*** (1.247)	3.286*** (1.197)	
tgt_fdi_total	-4.175 (4.673)	-0.963 (1.721)	-1.530 (1.836)	-5.755 (4.475)	
Bilateral FDI RRI distance					1.474 (1.683)
tgt_pmr_overall	1.470* (0.837)	-0.825*** (0.300)	-0.196 (0.249)	1.722* (0.896)	1.347 (0.832)
Observations	12758	13892	15293	14065	12758
Pseudo R-sq	0.489	0.477	0.455	0.472	0.489
Wald chi2	466.72	625.88	801.32	816.43	467.48
Prob χ^2 chi2	0.000	0.000	0.000	0.000	0.000

Standard errors in parentheses. OECD FDI Regulatory Restrictiveness Index (FDI RRI) measures cross-sectoral barriers to foreign direct investment. Range: 0 (fully open) to 1 (closed). Higher values = more restrictive. Col (1): 3-way FE (Year + Acq + Tgt). FDI RRI may identify if there is sufficient within-country policy reform variation. Col (2): Year + Acq FE only; target-country FDI RRI identifies off cross-target variation. Col (3): Year FE only; both acq and tgt FDI RRI identify (ELW-style identification). Col (4): Year + Tgt FE only; acquirer-country FDI RRI identifies off cross-acquirer variation. Col (5): 3-way FE with bilateral FDI RRI distance —acq - tgt—; tests regulatory-similarity friction. Table shows FDI RRI variables + tgt_pmr_overall and tgt_hc_openness_index (stability anchors); gravity controls suppressed. Pair clustering (acq x tgt) throughout. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 19: B13: PMR Decomposition (PPML, Year + Acq + Tgt FE)

	(1) Overall PMR Decomposed	(2) Pharmacy only	(3) Pharmacy only
tgt_pmr_overall	2.172** (1.002)		
tgt_pmr_state_distortions		-1.872 (1.202)	
tgt_pmr_entry_barriers		4.080*** (1.470)	
tgt_pmr_pharmacy		-0.284 (1.330)	1.046 (1.209)
tgt_pmr_retail_distribution		-0.684 (0.837)	
Observations	15097	15097	15097
Pseudo R-sq	0.500	0.501	0.499

Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B14 introduces the OECD iREG regulatory-governance pillars to the headline specification. Among the three pillars, only Regulatory Impact

Assessment (RIA) quality enters significantly: the RIA coefficient is positive and stable across the cascade (3.39, 4.16, and 4.98 in columns 1–3), while Stakeholder Engagement and Ex Post Evaluation are individually insignificant and add no incremental explanatory power (pseudo- R^2 unchanged at 0.506). Aggregate product-market restrictiveness (PMR) is rendered insignificant once RIA enters, indicating that on the OECD-coverage subsample the binding regulatory channel is the quality and predictability of the regulatory process rather than the level of restrictiveness. A one-standard-deviation increase in target-country RIA quality (0.17 on the 0–1 scale) is associated with an approximately 78 percent increase in expected inbound cross-border healthcare deal flow. Because RIA correlates most strongly, and negatively, with the PMR entry-barriers sub-index ($\rho = -0.30$), a natural concern is that RIA merely proxies low entry barriers. Table B14b addresses this directly: entered jointly under the full specification, the entry-barriers sub-index is significant in isolation (column 2) but loses significance once RIA is included, whereas RIA remains positive and significant (3.55, $p < 0.01$, column 3). The RIA coefficient is therefore stable (3.4–3.6) across specifications that introduce competing regulatory measures, supporting the interpretation that it captures an independent governance-quality channel distinct from regulatory restrictiveness. The bilateral-distance form (Table B14, column 4) is insignificant, but this reflects the further sample reduction to country-pairs with iREG coverage on both sides ($N = 7,647$); the better-powered target-level specification is preferred.

Table 20: B14: Regulatory Governance Channel (PPML, Year + Acq + Tgt FE)

	(1) +RIA	(2) +Stake	(3) +3 Pillars	(4) Distances
tgt_reg_ria	3.386*** (1.251)	4.157*** (1.613)	4.983** (1.938)	
tgt_reg_stakeholder		-0.874 (1.103)	-0.992 (1.095)	
tgt_reg_expost			-0.892 (1.250)	
Bilateral RIA distance				-1.161 (1.003)
Bilateral Stakeholder distance				-0.435 (0.708)
Bilateral Ex Post Eval. distance				-0.059 (0.379)
tgt_pmr_overall	0.266 (1.378)	0.790 (1.502)	0.762 (1.516)	0.408 (1.464)
Observations	12001	12001	12001	7647
Pseudo R-sq	0.506	0.506	0.506	0.486
Wald chi2	418.53	429.90	433.08	492.87
Prob χ^2	0.000	0.000	0.000	0.000

Standard errors in parentheses. Source: OECD Indicators of Regulatory Policy and Governance (iREG), 2014/2017/2021/2024 waves; LOCF fill applied. The three pillar scores (RIA, Stakeholder, Ex Post) are each aggregated from four cross-cutting categories (systematic adoption, methodology, oversight, transparency); the categories are therefore NOT entered jointly with the pillars to avoid mechanical collinearity. Composites are mean of binary indicator items within each pillar (Arndt et al. 2015). Cols 1-3: target-country pillar levels (acquirer side partly absorbed by acq FE). Col 4: bilateral absolute distances —acq - tgt— under 3-way FE (preferred form). Coverage: 36 percent of country-pair-years (OECD member countries only - flag as sensitivity sample). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 21: B14b: RIA vs PMR Entry Barriers Head-to-Head (PPML, Year + Acq + Tgt FE)

	(1) +RIA	(2) +Entry Barriers	(3) Both
tgt_reg_ria	3.367*** (1.243)		3.552*** (1.296)
tgt_pmr_entry_barr		2.634*** (0.997)	1.303 (1.309)
Observations	12001	15142	12001
Pseudo R-sq	0.506	0.499	0.506
Wald chi2	418.03	624.17	420.21
Prob χ^2 chi2	0.000	0.000	0.000

Standard errors in parentheses. Full headline specification (FULLSPEC) with tgt_pmr_overall replaced by the PMR entry-barriers sub-index ($\rho=0.93$ with overall PMR, so not entered jointly). Three-way FE: Year + acq_iso2 + tgt_iso2; pair clustering. OECD-coverage subsample (iREG). Col 3: entered jointly under the full specification, RIA remains positive and significant (+3.55, $p<0.01$) while entry barriers is insignificant; RIA is NOT displaced by entry barriers. RIA correlates only -0.30 with entry barriers (insufficient for absorption), confirming RIA captures regulatory process quality distinct from PMR restrictiveness rather than proxying entry barriers. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.